When the Network Moves the Currency: Bad Volatility Spillovers and Exchange Rate Pass-Through in Emerging Markets

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Abstract

When an emerging-market currency depreciates because the rest of the currency network is moving, do domestic prices respond as they would to an idiosyncratic depreciation of the same size? We measure each currency's exposure to the network with BadNet, the share of its depreciation-side (*bad*) volatility that is explained by shocks to other currencies and to global risk factors in a time-varying connectedness network of daily exchange rates. We then estimate panel local projections of consumer prices for Brazil, Chile, India, Mexico, South Africa, and Turkey over 2010–2026. Depreciations pass through to prices, but less so when BadNet is high: a one-standard-deviation increase in BadNet lowers the one-month pass-through of a 1% depreciation by 0.075 percentage points, about as much as the pass-through at the mean, and the reduction reaches 0.19 points after a year. The short-run result holds across alternative compositions of the network and, with a smaller magnitude, in a broader panel of fourteen inflation-targeting emerging markets. In advanced economies, where pass-through is small, BadNet makes little difference. The dampening is symmetric across depreciations and appreciations, comes from exposure to other emerging-market currencies, and is not explained by tighter monetary policy or financial conditions, which points to firms' uncertainty about the persistence of common movements.

Keywords: Exchange rate pass-through, Bad volatility, Currency networks, Emerging markets, Local projections
JEL classification: E31, F31, F41, G15, C33

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1. Introduction

The transmission of exchange rate movements to domestic prices has returned to the center of the international macroeconomic debate (Ha et al., 2020). The surge of inflation after the pandemic and the war in Ukraine coincided with the monetary tightening of 2022 and a broad appreciation of the dollar, which weighed on emerging-market currencies at a time when external cost pressures were already high (Obstfeld and Zhou, 2022; Carrière-Swallow et al., 2023), and central banks in these economies had to assess how much of each depreciation would end up in consumer prices. Such assessments are typically made in episodes in which many currencies fall at once. A large share of the variation of bilateral exchange rates against the dollar is common across currencies (Lustig et al., 2011; Verdelhan, 2018), and in our sample emerging-market currencies depreciate together far more often than they would if their movements were independent, with the largest depreciations concentrated in months in which most of them are falling at the same time.[2] This raises a question that the literature on exchange rate pass-through (ERPT) has not addressed, namely whether a depreciation that the domestic currency shares with other currencies reaches consumer prices in the same way as a depreciation of the same size that is specific to it.

To see why the question arises, and why it has remained open, it is useful to recall how the literature has explained the fact that the same exchange rate movement reaches prices with very different intensity across countries and over time. The first explanations located the source of these differences in the structure of markets. Exporters that price to market adjust their markups across destinations, and local distribution costs and nominal rigidities insulate consumer prices from changes in import costs, so that pass-through is incomplete and depends on the characteristics of each industry (Krugman, 1986; Dornbusch, 1987; Menon, 1995). This view implied that pass-through should be a fairly stable feature of each economy. The evidence of the 1990s contradicted that implication, since pass-through declined in many countries at roughly the same time (Taylor, 2000; Gagnon and Ihrig, 2004), and the debate moved from the structure of markets to the macroeconomic environment in which prices are set.

The most influential answer to this puzzle was given by Taylor (2000), who argued that firms pass through cost increases in proportion to how persistent they expect them to be, so that pass-through falls when inflation becomes low and stable and cost shocks are perceived as transitory. Cross-country evidence supported this view, showing that pass-through is positively associated with average inflation and that its decline coincided with a greater emphasis of central banks on inflation stabilization (Gagnon and Ihrig, 2004; Choudhri and Hakura, 2006), and models in which the frequency of price changes is endogenous provided a mechanism linking the inflation environment to pass-through (Devereux and Yetman, 2010). The macroeconomic explanation did not go unchallenged. Campa and Goldberg (2005) found that macroeconomic variables played only a minor role in the evolution of pass-through into import prices, and that shifts in the composition of imports were far more important, and the subsequent literature came to see the two explanations as complementary rather than competing (Burstein and Gopinath, 2014). For emerging economies, the main lesson of this phase was that monetary policy credibility

[2]Over 2010–2026, the six emerging-market currencies of our baseline sample depreciated against the dollar in the same month in 18.5% of the months, against 2.3% if their monthly movements had been independent with the same frequencies, and 80% of the monthly depreciations larger than 2% occurred in months in which at least four of the six were depreciating.

matters. Credible inflation-targeting frameworks keep inflation expectations anchored and are associated with lower pass-through, as the experience of South Africa illustrates, although pass-through in emerging economies remains higher than in advanced ones (López-Villavicencio and Mignon, 2017; Kabundi and Mlachila, 2019; Ha et al., 2020).

Yet similar depreciations continued to produce markedly different price responses within the same monetary regime (Forbes et al., 2018), which suggested that the determinants of pass-through must also lie in the exchange rate movement itself. This is where the current frontier of the literature stands. Shambaugh (2008) showed that consumer price pass-through is nearly complete in response to some structural shocks and low in response to others, and Forbes et al. (2018) and Ha et al. (2020) established that pass-through is relatively high after domestic monetary policy shocks and low after domestic demand shocks, with global shocks generating responses of their own, so that the inflationary content of a depreciation depends on its origin. At the same time, research on international pricing explained why the origin matters. Because most trade is invoiced in dollars, the dollar exchange rate rather than the bilateral rate governs import prices (Gopinath et al., 2020; Boz et al., 2022), and because firms' prices respond to those of their competitors, the effect of a depreciation on a firm's costs and prices depends on what happens to the currencies of other suppliers and competitors (Amiti et al., 2019). Taken together, the persistence argument and the pricing argument imply that the behavior of other currencies at the moment of a depreciation should matter for pass-through. When many currencies depreciate together, the relative cost position of domestic firms changes less than the bilateral rate suggests, and the movement, being part of a broader episode, is harder to read as permanent.

The empirical literature, however, has continued to treat each depreciation as an event of the domestic currency alone. Pass-through is estimated as the response of prices to the change in the domestic exchange rate, bilateral or trade-weighted (Campa and Goldberg, 2005; Burstein and Gopinath, 2014), and the origin of that change is either identified through restrictions in structural models estimated at low frequency (Shambaugh, 2008; Forbes et al., 2018; Ha et al., 2020) or associated with global factors such as the dollar, global risk, and U.S. monetary policy (Miranda-Agrippino and Rey, 2020; Obstfeld and Zhou, 2022). Effective exchange rates weight the currencies of trading partners, but they measure the movement of the domestic currency relative to them, not whether that movement is shared with them, and none of these approaches records whether, at the moment of a depreciation, the domestic currency is moving on its own or being carried by the movements of other currencies. Global factors measure how much global stress there is, not how strongly a particular currency is exposed to it, and currencies differ markedly in their exposure to common shocks (Lustig et al., 2011; Verdelhan, 2018). This omission is not innocuous. If most large depreciations in emerging economies are shared with other currencies, as the evidence above suggests, standard estimates of pass-through average over two economically different events, and the persistence and pricing arguments predict that these events reach prices with different intensity. The direction of the difference is itself an open question. A natural presumption is that shared depreciations, which typically occur in episodes of global stress, are especially inflationary, since they come with higher risk aversion, tighter external financing, and rising import costs, and depreciations against the dollar tighten financial conditions in emerging economies (Bruno and Shin, 2015; Hofmann et al., 2020; Obstfeld and Zhou, 2022). The persistence and pricing arguments point in the opposite direction.

Answering this question requires a measure of how much of a currency's movement

comes from other currencies that varies across currencies and over time, and such a measure can be built from the literature on financial connectedness. The approach of Diebold and Yilmaz (2009, 2012); Diebold and Yılmaz (2014) quantifies how much of the forecast error variance of each market is due to shocks originating in the others, and its extensions to time-varying parameters and frequency bands make it possible to follow the network of spillovers day by day (Baruník and Křehlík, 2018; Antonakakis et al., 2020). This literature has shown that volatility spillovers, limited in tranquil periods, intensify as crises unfold (Diebold and Yilmaz, 2012), a pattern that the contagion literature has taught to distinguish carefully from the high comovement that markets display in all periods (Forbes and Rigobon, 2002; Dungey and Gajurel, 2015). Building on the distinction between bad and good volatility, the volatility associated with adverse and favorable movements (Barndorff-Nielsen et al., 2010; Patton and Sheppard, 2015), it has also shown that spillovers are asymmetric and that, in the foreign exchange market, the asymmetries are dominated by bad rather than good volatility (Baruník et al., 2016, 2017). Connectedness measures have, however, been used mainly to describe financial networks, and when spillover methods have been brought to the study of pass-through, they have measured spillovers between exchange rates and prices directly (Balcilar and Usman, 2021). They have not been used to determine, for each currency and moment, whether a depreciation is the currency's own or the network's, which is the information the question above requires.

This study provides, to our knowledge, the first evidence on whether the pass-through of a depreciation depends on how much of it is shared with the international currency network. We estimate the conditional volatility of daily exchange rates with GJR-GARCH models (Glosten et al., 1993), retain the bad volatility, that is, the component associated with depreciation pressure, and estimate a time-varying connectedness network of twelve currencies, the broad dollar index, the VIX, and three economic policy uncertainty indices. From this network we obtain BadNet, the share of each currency's bad volatility that is explained by shocks to the other nodes, which is high when a currency is carried by the network and low when it moves on its own shocks. We then estimate panel local projections (Jordà, 2005) in which the response of consumer prices to depreciation varies with BadNet, for six emerging economies over 2010–2026, for a broader panel of fourteen inflation-targeting emerging markets, and for seven advanced economies (the euro area, Japan, Canada, the United Kingdom, Switzerland, Sweden, and Norway). Because month fixed effects absorb every shock common to the countries in the sample, the conditioning effect of BadNet is identified from differences across countries, within the same month, in the exposure of their currencies to the network.

The results contradict the presumption that shared depreciations are more inflationary. Depreciations remain inflationary in emerging economies, but their pass-through is substantially lower when the currency is carried by the network. In the six emerging economies, moving from the 25th to the 75th percentile of BadNet reduces the six-month pass-through of a 1% depreciation from 0.35 to 0.13 percentage points, a reduction of almost two-thirds, whereas no comparable effect appears in advanced economies, where pass-through is small at any level of network exposure. The association is smaller but present in the broader emerging panel, and it is specific to bad volatility, since networks built from total volatility do not reproduce it. Consistent with the persistence argument, the dampening is symmetric between depreciations and appreciations, comes from exposure to other emerging-market currencies, is stronger when inflation is already high, and is not accompanied by larger policy-rate increases or tighter financial conditions.

These findings contribute to the literature in three ways. First, they identify a new determinant of pass-through, the extent to which a depreciation is shared with the currency network, and show that it is quantitatively first-order. To the explanations of why the same exchange rate movement reaches prices with different intensity, namely the structure of markets, the inflation regime, and the structural shock behind the movement, they add a dimension that varies from month to month within the same country and regime, and that structural decompositions do not capture. Second, they show that the direction of this effect is the opposite of the common presumption. Shared depreciations, typical of episodes of global stress, pass through less rather than more, which implies that standard pass-through estimates mix two different events and suggests one reason why pass-through appears unstable within monetary regimes (Forbes et al., 2018). Third, they provide a measure of this state that is market-based, currency-specific, available at daily frequency, and free of structural identifying restrictions, and the evidence on channels links the result to the persistence mechanism proposed by Taylor (2000), which here operates across currencies through the difficulty of reading movements shared with the network. For central banks in emerging economies, the results imply that the inflationary risk of a depreciation depends not only on its size but also on whether it occurs in isolation or together with other currencies, a distinction that can be monitored in real time.

The remainder of the article is organized as follows. Section 2 develops the literature in more detail and sets out the channels through which network exposure may affect pass-through. Section 3 describes the construction of BadNet and the empirical strategy. Section 4 presents the results, and Section 5 concludes.

2. Literature Review and Empirical Question

2.1. Exchange Rate Pass-Through as a Conditional Phenomenon

The introduction traced how the explanation of pass-through moved from the structure of markets to the inflation environment and, more recently, to the origin of exchange rate movements. This section adds the elements of that literature on which the empirical design and the conceptual framework of Section 2.4 rely. The first concerns the microeconomics of price setting. Whether a depreciation reaches import and consumer prices depends on the currency in which prices are set (Betts and Devereux, 1996; Gopinath et al., 2010), and because most international trade is invoiced in dollars, the dollar exchange rate matters for import prices even when the United States is not a party to the transaction (Goldberg and Tille, 2008; Gopinath et al., 2020; Boz et al., 2022). It also depends on how firms respond to their competitors. Amiti et al. (2019) show that firms' prices respond strongly to the prices of their competitors and that this strategic complementarity is concentrated among large firms, which implies that the effect of a cost shock on a firm's price depends on whether the same shock also hits its competitors.

The second concerns emerging economies. Pass-through is higher in emerging than in advanced economies and is positively related to the level of inflation (Choudhri and Hakura, 2006; Ha et al., 2020). The adoption of inflation targeting and the gains in monetary policy credibility since the 2000s have lowered it, by keeping inflation expectations anchored when the exchange rate moves (Mishkin and Schmidt-Hebbel, 2007; López-Villavicencio and Mignon, 2017; Kabundi and Mlachila, 2019), but the evidence for emerging and transition economies shows that pass-through remains economically relevant and varies considerably across countries and periods (Ben Cheikh and Ben Zaied, 2020; Nasir et al., 2020; Pham et al., 2023). The third concerns the form of the relationship. Pass-through differs between

depreciations and appreciations and varies with the size of exchange rate changes (Delatte and López-Villavicencio, 2012; Bussière, 2013; Ben Cheikh, 2012; Anderl and Caporale, 2023; Lama et al., 2025), and it increases with the volatility of the exchange rate (Alexius and Holmberg, 2024). These findings motivate two features of our design, the use of the depreciation component of exchange rate changes as the main regressor and the controls for the size and own volatility of depreciations, which ensure that the network state is not confused with the nonlinearity of large movements.

2.2. Global Shocks, Dominant Currency, and Imported Inflation

Global shocks reach domestic inflation through several channels besides the exchange rate. Commodity, energy, and shipping cost shocks raise import and production costs directly (Abbas and Lan, 2020; Carrière-Swallow et al., 2023; Mirza et al., 2023), and the global financial cycle transmits U.S. monetary policy to asset prices, capital flows, and credit conditions across countries (Miranda-Agrippino and Rey, 2020). A related literature shows that the exchange rate itself operates as a financial channel in emerging economies. A depreciation against the dollar tightens financial conditions by raising the domestic value of dollar debt, reducing cross-border bank lending, and widening local-currency bond risk premia (Bruno and Shin, 2015; Hofmann et al., 2020), and the dollar cycle of the past decade has made these effects especially visible (Obstfeld and Zhou, 2022). This channel can push pass-through in either direction. Tighter financing raises costs and may unanchor expectations, which would amplify pass-through, but it also compresses demand, which would dampen it. Section 2.4 treats both possibilities explicitly.

The literature on currency risk adds a further element. A large share of the variation of bilateral exchange rates against the dollar is driven by common factors, but currencies differ markedly in their loadings on these factors (Lustig et al., 2011; Verdelhan, 2018). Two emerging economies facing the same global shock may therefore see their currencies respond to very different degrees. Common indicators of global conditions, such as the VIX or the dollar index, capture the shock but not the exposure of each currency to it, and they enter the pass-through literature as controls or as identifying variables rather than as measures of how much of a given currency's movement is shared with others. Our measure is designed to capture that exposure.

2.3. Currency Networks and Bad Volatility

The connectedness approach of Diebold and Yilmaz (2009, 2012); Diebold and Yılmaz (2014) measures, from the forecast error variance decomposition of a vector autoregression, how much of the variance of each variable is due to shocks to the others. Summing these shares across sources gives the connectedness that each market receives, and summing them across destinations gives the connectedness that it transmits, so that the approach identifies the transmitters and receivers of shocks in a system. Baruník and Křehlík (2018) decompose these measures by frequency, which separates short-lived spillovers from persistent ones, and time-varying parameter VARs estimated with the Kalman filter allow the network to change at each date without the arbitrary choice of a rolling window (Koop and Korobilis, 2014; Antonakakis et al., 2020). This methodology complements the literature on financial contagion. Forbes and Rigobon (2002) show that much of the apparent increase in cross-market correlations during crises reflects the high interdependence that markets display in all periods, and later work has refined the conditions under which transmission during crises exceeds what fundamentals explain (Bekaert et al., 2005; Dungey and Gajurel, 2015; Gajurel and Thapa, 2026). Connectedness

measures do not settle this debate, but they describe, period by period, how much of each market's variance comes from the others.

Applied to currencies, the approach has an additional dimension. Building on the evidence that negative and positive returns carry different information (Barndorff-Nielsen et al., 2010; Patton and Sheppard, 2015), Baruník et al. (2016, 2017) separate spillovers of bad and good volatility and find that, in the foreign exchange market, the asymmetries in spillovers are dominated by bad volatility. For an importing economy, the relevant side of exchange rate risk is depreciation, and the measure we need is the connectedness that a currency receives in the network of bad volatility, the share of its depreciation-side volatility that is explained by shocks to other currencies and to global risk factors. This measure, which we call BadNet, is not a measure of how volatile a currency is, nor of how much global risk there is. It measures how much of a currency's depreciation-side volatility comes from the network.

2.4. Conceptual Framework: How Network Exposure Can Change Pass-Through

Received connectedness, $\text{FROM}^{B}_{i,t}$, is high when most of the bad volatility of currency i is explained by shocks to other nodes and low when it is driven by the currency's own shocks. A depreciation of a given size therefore has a different composition in the two states. Under low BadNet it is mostly idiosyncratic; under high BadNet it is mostly the domestic counterpart of a movement shared with other currencies and with global risk factors. The literature reviewed above suggests at least four channels through which this difference in composition could change the inflationary effect of the same depreciation. They do not all predict the same sign, which is why the specification does not impose one.

The first channel is uncertainty about persistence. In Taylor (2000), firms pass through cost shocks in proportion to their expected persistence. Theories of price setting under imperfect information add that firms do not observe the nature of shocks directly and respond less to those that are harder to interpret. In the model of rational inattention of Maćkowiak and Wiederholt (2009), firms allocate their limited attention to the conditions specific to their own market and therefore adjust prices quickly to idiosyncratic shocks but slowly to aggregate ones. A depreciation that the currency shares with the rest of the network is, from the point of view of a domestic firm, closer to an aggregate shock than an idiosyncratic one, and it is harder to read, because it may reflect a global risk episode that unwinds or a lasting change in external conditions. The channel does not require common movements to reverse more often than idiosyncratic ones, only that their persistence be harder to assess when they occur. It predicts a lower pass-through under high BadNet for depreciations and appreciations alike, and a stronger dampening when inflation is already high, because firms that have recently raised prices have less reason to react to a movement whose persistence is uncertain.

The second channel is the effective exchange rate and competitors' prices. With strategic complementarities in price setting, a firm's optimal price depends on the prices of its competitors (Amiti et al., 2019), and import costs depend on the currencies of all suppliers. When the depreciation against the dollar is part of a common movement of emerging-market currencies, the depreciation against competitors and suppliers in other emerging economies is smaller than against the dollar, and so is the change in relative prices. This channel predicts a negative interaction that is weaker where imports are invoiced in dollars, since under dominant currency pricing the dollar rate is the relevant one for import prices (Gopinath et al., 2020; Boz et al., 2022); stronger in economies that are more central in the trade network and therefore more exposed to common shocks

(Richmond, 2019); driven by exposure to the other emerging currencies rather than to global nodes; and smaller when depreciation is measured in nominal effective terms.

The third channel is demand and financial contraction. Through the financial channel of the exchange rate, depreciations against the dollar tighten financial conditions in emerging economies (Bruno and Shin, 2015; Hofmann et al., 2020), and high-BadNet depreciations may therefore coincide with capital outflows, wider sovereign spreads, tighter credit, and weaker activity (Miranda-Agrippino and Rey, 2020), which reduce firms' ability to raise prices. This channel predicts a negative interaction concentrated in depreciations, together with a tightening of financial conditions and a fall in activity after high-BadNet depreciations. The fourth channel is a defensive monetary response. If central banks raise rates more when depreciation occurs under network stress, the pass-through would fall because policy offsets it (Calvo and Reinhart, 2002). This channel predicts larger policy-rate increases after high-BadNet depreciations.

The opposite sign is also possible, and it corresponds to the presumption discussed in the introduction. If high BadNet signals deteriorating external conditions that raise financing costs and unanchor expectations (Hofmann et al., 2020; Obstfeld and Zhou, 2022), it would amplify pass-through and the interaction would be positive. Table 1 summarizes the mechanisms and their testable predictions, which Section 4 confronts with the data. The empirical question is therefore whether exposure to BadNet is associated with a different marginal effect of depreciation and, if so, which of these channels its pattern is consistent with. Because the design does not identify exogenous exchange rate shocks, the evidence discriminates among channels through their testable implications rather than by isolating any one of them.

Table 1: Channels through which BadNet could change pass-through: mechanisms and testable predictions

Channel	Mechanism	Testable prediction	Sign
Uncertainty about persistence	Firms respond less to shocks whose persistence is hard to assess, and shared movements are closer to aggregate shocks (Taylor, 2000; Maćkowiak and Wiederholt, 2009)	Lower pass-through of both depreciations and appreciations; stronger dampening when inflation is high	−
Effective exchange rate and competitors' prices	Shared depreciations change relative prices against competitors and suppliers less (Amiti et al., 2019; Gopinath et al., 2020)	Weaker dampening with dollar invoicing; stronger with trade centrality; driven by exposure to other emerging currencies; smaller in effective terms	−
Demand and financial contraction	Depreciation tightens financing and compresses demand (Bruno and Shin, 2015; Hofmann et al., 2020)	Dampening concentrated in depreciations; wider spreads, less credit, lower activity	−
Defensive monetary policy	Central banks offset network-driven depreciations (Calvo and Reinhart, 2002)	Larger policy-rate increases after high-BadNet depreciations	−
Amplification	Shared depreciations signal external stress and unanchor expectations (Obstfeld and Zhou, 2022)	Higher pass-through under high BadNet	+

Note: The sign refers to the predicted sign of the interaction between depreciation and BadNet. The channels are not mutually exclusive. Section 4.11 confronts these predictions with the evidence.

3. Empirical Strategy

3.1. Overview of the Empirical Strategy

The empirical strategy has two stages. The first measures, for each currency and day, how much of its bad volatility comes from the rest of the currency network (Section 3.2). The second asks whether this exposure changes the pass-through of depreciation to consumer prices, using panel local projections (Sections 3.3 to 3.5). Section 3.6 describes the samples, and Sections 3.7 and 3.8 discuss inference and identification.

BadNet is not a proxy for global risk. The VIX, policy uncertainty, and the dollar measure the level of common stress, and month fixed effects absorb them. BadNet measures how much of a given currency's bad volatility comes from shocks elsewhere in the network, which differs across currencies that face the same global shock. The baseline measure is the currency's received connectedness in the bad-volatility network; the aggregate connectedness index, the average exposure of the other emerging currencies, the connectedness the currency transmits and its net position, networks of total volatility, frequency components, the good-volatility network, and alternative network compositions are used as alternatives.

3.2. Measuring Bad Volatility and Constructing BadNet

The construction of BadNet follows the idea that aggregate volatility may conceal relevant differences between favorable and adverse movements (Barndorff-Nielsen et al., 2010; Patton and Sheppard, 2015). In foreign exchange markets, this distinction requires particular care because the statistical sign of returns depends on the quotation convention. In this article, every node is oriented so that an increase indicates pressure that is unfavorable from the perspective of an emerging-market importer. Bilateral exchange rates are measured in units of domestic currency per U.S. dollar, so that increases in $E_{i,t}$ indicate a depreciation of the domestic currency; the euro and the Australian dollar, which are quoted in dollars per unit, are inverted. The U.S. dollar node is the Federal Reserve broad trade-weighted dollar index, whose increases indicate a stronger dollar and therefore depreciation pressure on the rest of the system. For the VIX and the economic policy uncertainty indices, the return is the daily log change of the index, so that positive values indicate rising risk aversion or uncertainty. Depreciation movements are thus treated as adverse from a domestic macroeconomic perspective, since they raise import costs and inflationary pressures and tend to tighten external financial conditions in emerging economies (Ha et al., 2020; Hofmann et al., 2020).

For each node i, the daily return is defined as:

$$r_{i,t} = 100 \times \left[\ln(E_{i,t}) - \ln(E_{i,t-1})\right], \tag{1}$$

where $E_{i,t}$ denotes the nominal exchange rate or the level of the risk factor. The decomposition between good and bad volatility is obtained from GJR-GARCH(1,1) models (Glosten et al., 1993), which capture the asymmetric response of conditional variance to shocks of different signs.

For a return series r_t, the mean equation is an ARMA(1,1):

$$r_t = \mu + \phi r_{t-1} + \vartheta \varepsilon_{t-1} + \varepsilon_t, \tag{2}$$

where ε_t is the innovation of the series. We assume that:

$$\varepsilon_t = z_t \sigma_t, \tag{3}$$

where σ_t^2 is the conditional variance and z_t is a standardized innovation with zero mean and unit variance, drawn from a Student-t distribution to accommodate the fat tails of daily exchange rate returns. The conditional variance follows a GJR-GARCH(1,1) specification, which lets the variance respond asymmetrically to positive and negative shocks:

$$\sigma_t^2 = \omega + \alpha\varepsilon_{t-1}^2 + \gamma I_{\{\varepsilon_{t-1}<0\}}\varepsilon_{t-1}^2 + \beta\sigma_{t-1}^2, \tag{4}$$

where $I_{\{\varepsilon_{t-1}<0\}}$ is an indicator function associated with negative shocks under the standard statistical convention of the model. Since, under our quotation convention, positive innovations correspond to depreciation, a negative γ indicates that conditional variance rises more after depreciation than after appreciation. The estimated γ is negative and statistically significant for most currencies and for the VIX (Table A1 in the Appendix), which supports treating the depreciation side as the bad-volatility layer.

The decomposition between *good volatility* and *bad volatility* follows the logic of the semivariance literature, according to which positive and negative movements carry different informational content (Barndorff-Nielsen et al., 2010; Patton and Sheppard, 2015; Baruńík et al., 2016). In our implementation, the split is applied to the GJR-GARCH conditional variance according to the sign of the standardized innovation:

$$GV_t = \sigma_t^2 \, I_{\{z_t \leq 0\}}, \tag{5}$$

$$BV_t = \sigma_t^2 \, I_{\{z_t > 0\}}. \tag{6}$$

Bad volatility is assigned to positive innovations, the reverse of the usual convention for asset returns, because under our orientation positive innovations correspond to depreciation, rising risk aversion, rising uncertainty, or a stronger dollar. Using conditional variances rather than squared returns keeps both layers strictly positive on the days they are active and lets the volatility persistence estimated by the GJR-GARCH model carry over to the network stage. Across exchange rates, the share of days of depreciation pressure lies between 0.43 and 0.51. Each layer is therefore zero on roughly half of the days, and the series that enter the network are intermittent. The connectedness measures should accordingly be read as describing how conditional variance on days of depreciation pressure co-moves across nodes, not as a model of a continuous latent process; the monthly averaging of BadNet attenuates the day-to-day intermittency. Throughout the article, *bad volatility* denotes BV_t, the conditional variance on days of depreciation pressure, and *good volatility* denotes GV_t; the networks estimated from them are referred to as the bad-volatility and good-volatility networks.

Two nodes require special treatment. Daily economic policy uncertainty indices are not available for China and Russia, and we use the monthly indices of Baker et al. (2016) repeated across the business days of each month. For these two series the GJR-GARCH estimates are degenerate ($\beta \approx 0.001$, $\gamma \approx 0.97$) and the bad-volatility layer is active on 91% (China) and 83% (Russia) of days, a mechanical consequence of a step function whose only innovations occur at the turn of the month. Because these nodes could shape the persistence of BadNet, Section 4.7 reports the main results for a network without them and for a network that replaces them with daily market-based proxies, the CBOE China ETF volatility index (VXFXI) and the Russian volatility index (RVI), over the period in which both are available.

Using the *bad volatility* series, we estimate a dynamic network of bad-volatility transmission. Let $\mathbf{y}_t$ be an N-dimensional vector containing the bad volatilities of currencies

and global risk factors. This vector is represented by a time-varying parameter VAR model:

$$\mathbf{y}_t = \sum_{j=1}^{p} \mathbf{\Phi}_{j,t} \mathbf{y}_{t-j} + \mathbf{u}_t, \tag{7}$$

where $\mathbf{\Phi}_{j,t}$ are time-varying coefficient matrices and $\mathbf{u}_t$ is a vector of innovations with variance-covariance matrix $\mathbf{\Sigma}_t$. The main advantage of this specification is that it allows the transmission structure among currencies and global risk factors to evolve over time, which is essential in a sample marked by financial crises, the pandemic, geopolitical shocks, and shifts in the international monetary cycle. We follow Antonakakis et al. (2020), who adapt the forgetting-factor approach of Koop and Korobilis (2014): the model has $p = 2$ lags, the Kalman filter is initialized with a prior obtained from a constant-parameter VAR estimated on the first 200 observations, and the forgetting and decay factors are both set to 0.99. Forecast error variance decompositions are computed at horizon $H = 10$ trading days.

From the TVP-VAR representation, we compute generalized forecast error variance decompositions, which do not depend on the ordering of the variables (Koop et al., 1996; Pesaran and Shin, 1998). Let $\theta_{ij,t}(H)$ denote the share of the forecast error variance of variable i explained by shocks to variable j, at forecast horizon H. Since the elements of the generalized decomposition do not necessarily sum to one across each row, we use the normalized version:

$$\widetilde{\theta}_{ij,t}(H) = \frac{\theta_{ij,t}(H)}{\sum_{j=1}^{N} \theta_{ij,t}(H)}. \tag{8}$$

Based on this matrix, we compute the total connectedness index of the bad-volatility network:

$$\mathrm{TCI}_t^B(H) = \frac{\sum_{i,j=1, i\neq j}^{N} \widetilde{\theta}_{ij,t}(H)}{N} \times 100. \tag{9}$$

The TCI_t^B measures the systemic intensity of bad-volatility transmission in the global foreign exchange market. Higher values indicate a larger contribution of cross-market shocks to forecast error variance, suggesting greater integration of the bad-volatility network.

In addition to the aggregate index, we compute directional measures of currency-specific exposure. The connectedness received by currency i from the remaining nodes is given by:

$$\mathrm{FROM}_{i,t}^B(H) = \sum_{j=1, j\neq i}^{N} \widetilde{\theta}_{ij,t}(H) \times 100, \tag{10}$$

BadNet is $\mathrm{FROM}_{i,t}^B$, the share of the forecast error variance of currency i's bad volatility that is due to shocks to the other nodes. It is high when the network moves the currency and low when the currency moves on its own shocks. The aggregate index TCI_t^B, which measures the state of the network as a whole, is used as an alternative measure, as are the connectedness that currency i transmits to the other nodes,

$$\mathrm{TO}_{i,t}^B(H) = \sum_{j=1, j\neq i}^{N} \widetilde{\theta}_{ji,t}(H) \times 100, \tag{11}$$

and its net position, $\text{NET}^B_{i,t} = \text{TO}^B_{i,t} - \text{FROM}^B_{i,t}$. To check that the information lies in the depreciation side of volatility, we also build two networks of total volatility with the same TVP-VAR, one from the conditional variance of a symmetric sGARCH(1,1) model with Student-t errors and one from the full GJR-GARCH variance $\sigma_t^2 = GV_t + BV_t$.

The frequency-domain approach of Baruník and Křehlík (2018) allows BadNet to be decomposed into short-run and persistent components. The short-run component captures rapid market reactions, liquidity adjustments, portfolio rebalancing, and transitory transmission of bad-volatility shocks across currencies. The persistent component captures longer-lasting states associated with global financing conditions, risk premia, external vulnerability, and prolonged exchange rate stress regimes. In the local projections, these components are separately incorporated as $BadNet^{short}_{i,t}$ and $BadNet^{persistent}_{i,t}$, allowing us to assess whether ERPT conditioning arises from transitory network shocks, persistent network states, or both.

In the empirical implementation, the spectral band is partitioned at the frequency $\pi/2$. The short-run component collects spillovers at frequencies in $(\pi/2, \pi]$ and the persistent component those in $[0, \pi/2]$. Following the labeling convention common in applications of Baruník and Křehlík (2018), we refer to the band $(\pi/k, \pi]$ as cycles of up to k trading days; strictly, it contains cycles with periods between 2 and $2k$ days. This classification is appropriate for daily exchange rate data because spillovers transmitted within a few trading days capture immediate financial-market adjustments, such as news absorption, liquidity shocks, portfolio rebalancing, and rapid risk repricing. Spillovers at lower frequencies, by contrast, contain information about more persistent states of global currency stress, systemic exposure, and macro-financial fragility. Under this convention, the cut at $\pi/22$ separates cycles of up to 22 trading days, which strictly includes periods of up to 44 trading days, about two months. Because a forecast horizon of ten days yields a coarse frequency grid, the sensitivity of the split to the cut-off is assessed with $H = 100$ and cut-offs at $\pi/2$, $\pi/5$, and $\pi/22$.

To incorporate BadNet into the inflation models, the daily connectedness series are aggregated to the monthly frequency. The baseline specification uses the monthly average of BadNet, which captures the average state of the network during the price formation period. Alternative measures, such as the monthly maximum and the end-of-month value, are used in robustness analyses. To make coefficients comparable across specifications, monthly BadNet is standardized using the mean (53.45) and standard deviation (16.90) of the emerging-economy panel, so that $BadNet_{i,t}$ is expressed in standard deviations from the emerging-market mean. Appendix Figure A1 plots the monthly series for the six emerging economies.

3.3. Local Projections for BadNet-Conditioned Pass-Through

The second empirical stage estimates BadNet-conditioned ERPT using panel local projections, following Jordà (2005). This approach is suitable because it allows us to estimate the inflation response directly at different horizons without imposing a single dynamic structure on the entire transmission path (Plagborg-Møller and Wolf, 2021). Moreover, local projections facilitate the interpretation of the marginal effect of currency depreciation under different states of the bad-volatility currency network.

The main dependent variable is the cumulative change in consumer prices between the month before the depreciation and horizon h:

$$\pi^c_{i,t+h} = 100 \times \left[\ln(CPI_{i,t+h}) - \ln(CPI_{i,t-1})\right], \tag{12}$$

where $CPI_{i,t}$ denotes the consumer price index of country i. With this definition, the coefficient on depreciation at horizon h is the cumulative pass-through of a 1% depreciation, in percentage points, which is the standard object in the ERPT literature. Year-on-year inflation at $t+h$ would overlap with up to eleven months before the depreciation; results with that variable, with monthly inflation at $t+h$, and with core and sectoral price indices are reported as robustness.

The baseline specification is given by:

$$\begin{aligned}\pi^{c}_{i,t+h} = \mu_i + \tau_t + \beta_1 \Delta e^{+}_{i,t} + \beta_2 BadNet_{i,t} \\ + \beta_3 \left(\Delta e^{+}_{i,t} \times BadNet_{i,t}\right) + \boldsymbol{\Theta}' \mathbf{X}_{i,t} + \xi_{i,t+h},\end{aligned} \tag{13}$$

for $h = 0, 1, \ldots, 12$ (the forecast horizon of the inflation response, distinct from the horizon H of the variance decomposition); tables report $h = 0, 1, 3, 6, 12$. The term $\Delta e^{+}_{i,t}$ denotes the depreciation component of exchange rate changes, defined as:

$$\Delta e^{+}_{i,t} = \max(\Delta e_{i,t}, 0), \tag{14}$$

where:

$$\Delta e_{i,t} = 100 \times \left[\ln(E_{i,t}) - \ln(E_{i,t-1})\right]. \tag{15}$$

The use of $\Delta e^{+}_{i,t}$ reflects the asymmetric nature of ERPT. The inflationary channel is expected to operate primarily during episodes of domestic currency depreciation, since such episodes raise import prices, input costs, and the prices of fuel and tradable food, and may affect inflation expectations. Appreciations may reduce import prices, but their effects tend to be smaller, slower, or partially retained in firms' margins, as suggested by the literature on pass-through asymmetry (Delatte and López-Villavicencio, 2012; Bussière, 2013; Pham et al., 2023; Lama et al., 2025). A specification with separate depreciation and appreciation terms is reported in Section 4.10.

The term $BadNet_{i,t}$ represents the domestic currency's exposure to the global network of *bad volatility*. In the baseline specification, we use the standardized received connectedness of the domestic currency in the bad-volatility network, $\text{FROM}^{B}_{i,t}$, because this measure captures the country's exposure to bad-volatility shocks originating in the rest of the system. In alternative specifications, we use the aggregate index TCI^{B}_{t}, the average exposure of the other emerging currencies, BadNet frequency components, and the corresponding measure based on *good volatility*.

The vector $\mathbf{X}_{i,t}$ contains three lags of monthly inflation, of $\Delta e^{+}_{i,t}$, of $BadNet_{i,t}$, and of $\Delta e^{+}_{i,t} \times BadNet_{i,t}$. The country fixed effects μ_i and month fixed effects τ_t are included in every specification. The month fixed effects absorb every variable that is common to all countries in a given month, including the VIX, U.S. policy and long-term interest rates, the broad dollar, oil, food, and commodity prices, and international transportation costs, so these global controls cannot enter separately. Domestic variables that may respond to the depreciation within the same horizon, such as the policy rate and economic activity, are not included in the baseline to avoid conditioning on outcomes; specifications that add the lagged policy rate and interactions of depreciation with the VIX, the broad dollar, and a risk-off indicator are reported in Section 4.6.

The coefficient β_1 measures the average effect of currency depreciation on inflation at horizon h when BadNet is at the emerging-market mean, conditional on the remaining controls. The coefficient β_2 captures the direct association between BadNet and domestic inflation. The coefficient β_3 is the central parameter of the article, as it indicates whether

exposure to the bad-volatility network is associated with a different marginal effect of currency depreciation on inflation. Because $BadNet_{i,t}$ is standardized, β_3 is the change in the cumulative pass-through of a 1% depreciation, in percentage points, associated with a one-standard-deviation increase in BadNet.

The interpretation of β_3 does not impose a one-sided hypothesis. Positive values indicate that BadNet intensifies ERPT, suggesting that depreciations are more inflationary when the currency is more exposed to the bad-volatility network. Negative values indicate that BadNet reduces the marginal effect of depreciation, through one or more of the channels of Section 2.4. Values statistically indistinguishable from zero indicate that the bad-volatility network does not meaningfully alter pass-through at the horizon under analysis.

The marginal interpretation of BadNet-conditioned pass-through is given by:

$$\frac{\partial \pi^{c}_{i,t+h}}{\partial \Delta e^{+}_{i,t}} = \beta_1 + \beta_3 BadNet_{i,t}. \tag{16}$$

Thus, ERPT is not modeled as a fixed elasticity. It varies according to whether the domestic currency is more or less exposed to the international network of bad-volatility transmission.

3.4. Heterogeneity between Emerging and Advanced Economies

To assess whether the conditioning effect of BadNet differs between emerging and advanced economies, we estimate a specification with a triple interaction:

$$\begin{aligned} \pi^{c}_{i,t+h} = \mu_i + \tau_t &+ \beta_1 \Delta e^{+}_{i,t} + \beta_2 BadNet_{i,t} \\ &+ \beta_3 \left(\Delta e^{+}_{i,t} \times BadNet_{i,t}\right) \\ &+ \beta_4 \left(\Delta e^{+}_{i,t} \times Emerging_i\right) \\ &+ \beta_5 \left(BadNet_{i,t} \times Emerging_i\right) \\ &+ \beta_6 \left(\Delta e^{+}_{i,t} \times BadNet_{i,t} \times Emerging_i\right) \\ &+ \mathbf{\Theta}' \mathbf{X}_{i,t} + \xi_{i,t+h}. \end{aligned} \tag{17}$$

The variable $Emerging_i$ equals one for emerging economies and zero for advanced economies. The specification is estimated with the three advanced economies of the main network and, as a check on the power of the comparison, with the expanded group of seven advanced economies described in Section 3.6. The advanced-economy group serves as the reference category for assessing whether the conditioning effect of BadNet is specific to economies that are more vulnerable to exchange rate movements or whether it reflects a common pattern across country groups. The coefficient β_6 measures whether the conditioning effect of BadNet on ERPT differs in emerging economies. The marginal effect of BadNet on pass-through in advanced economies is given by β_3, while the corresponding effect in emerging economies is given by $\beta_3 + \beta_6$. In this specification BadNet is standardized with the same emerging-market moments as in the baseline, so that $\beta_3 + \beta_6$ is directly comparable with the baseline β_3.

The distinction between emerging and advanced economies is relevant because emerging economies are more sensitive to capital flows, global liquidity, and changes in the monetary cycle of core economies (Bruno and Shin, 2015; Miranda-Agrippino and Rey, 2020), and because their pass-through is higher (Choudhri and Hakura, 2006; Ha et al., 2020). In addition, currency depreciations in emerging economies may carry information

about deteriorating expectations, inflationary pressures, higher external financing costs, and balance-sheet fragilities in foreign currency. For this reason, if the state of the bad-volatility network matters for pass-through, its effect should be most visible in these countries.

The models are estimated for different horizons h, allowing us to assess the time profile of exchange rate transmission. Short horizons capture immediate effects on imported goods, fuel, food, and tradable components of inflation. Longer horizons capture propagation to consumer prices, expectations, contracts, wage adjustments, administered prices, and non-tradable components. This structure is consistent with the literature that treats ERPT as a dynamic, incomplete, and horizon-dependent process (Campa and Goldberg, 2005; Burstein and Gopinath, 2014; Ha et al., 2020; Alexius and Holmberg, 2024).

3.5. Restrictive Monetary Regimes and Frequency Decomposition

Beyond the comparative specification between emerging and advanced economies, we estimate a regime specification to assess whether the conditioning effect of BadNet changes in environments of greater monetary restriction. Because policy-rate levels differ widely across countries (a few high-rate economies would account for most observations above a pooled threshold), the regime is defined within each country. In the baseline, $\text{HighPolicy}_{i,t}$ is an indicator variable equal to one when the domestic policy rate in $t-1$ is at or above the country's own sample median. We also report results for the country's upper quartile and for the policy rate standardized within country and entered continuously. The specification includes interactions among $\Delta e^{+}_{i,t}$, $\text{BadNet}_{i,t}$, and $\text{HighPolicy}_{i,t}$:

$$\begin{aligned}\pi^{c}_{i,t+h} = \mu_i + \tau_t &+ \delta_1 \Delta e^{+}_{i,t} + \delta_2 BadNet_{i,t} + \delta_3 HighPolicy_{i,t} \\ &+ \delta_4 \left(\Delta e^{+}_{i,t} \times HighPolicy_{i,t}\right) \\ &+ \delta_5 \left(\Delta e^{+}_{i,t} \times BadNet_{i,t}\right) \\ &+ \delta_6 \left(\Delta e^{+}_{i,t} \times BadNet_{i,t} \times HighPolicy_{i,t}\right) \\ &+ \boldsymbol{\Theta}' \mathbf{X}_{i,t} + \xi_{i,t+h}.\end{aligned} \tag{18}$$

The purpose of this specification is not to identify an isolated causal monetary policy channel, but to assess whether BadNet is more informative when currency depreciation occurs in high-interest-rate states, in which inflationary pressure is typically higher and monetary conditions tighter. The coefficient of interest is δ_6, which indicates whether the conditioning effect of BadNet changes under these regimes; δ_3 and δ_4 are included so that δ_6 is not confounded with level or slope differences between regimes. The same structure is used with indicators for inflation above the country median, exchange rate volatility above the country median, and private credit growth above the country median, all measured in $t-1$.

The frequency decomposition is incorporated by replacing aggregate BadNet with its short-run and persistent components, each standardized. For each component $k \in \{short, persistent\}$, we estimate equation (13) with $BadNet^{k}_{i,t}$ in place of $BadNet_{i,t}$ and denote the interaction coefficient by λ_3. This specification allows us to assess whether ERPT conditioning arises from transitory network shocks, persistent states of systemic exposure, or both. Because the comparative specification shows that the conditioning effect is confined to emerging economies, the frequency decomposition is estimated on the emerging-economy panel, where λ_3 is directly comparable with β_3.

3.6. Sample, Temporal Aggregation, and Variable Treatment

The empirical sample differs across the two stages of the analysis. The first stage constructs the global currency network, while the second stage estimates exchange rate pass-through (ERPT) to domestic inflation. This distinction is necessary because BadNet is intended to capture the global network of bad-volatility transmission, whereas the ERPT regressions require macroeconomic comparability across countries.

In the network stage, the daily currency system includes major currencies, emerging-market currencies, and global risk and uncertainty factors. The main network has seventeen nodes: the Brazilian real, Mexican peso, Chilean peso, South African rand, Turkish lira, Indian rupee, Chinese yuan, Korean won, euro, Japanese yen, Canadian dollar, Australian dollar, the broad U.S. dollar index, the VIX, and the economic policy uncertainty indices for the United States, China, and Russia. The daily sample runs from 3 January 2006 to 1 June 2026, with 5,325 business days; missing values due to national holidays are filled with the previous observation. The data dictionary, with sources and transformations, is reported in Table A2 in the Appendix. The Chilean peso is the observed dollar rate of the Banco Central de Chile, and the Chinese yuan enters as a separate node. The dollar, the yuan, and the won are retained in the network because they are central nodes of global currency transmission, even though the United States, China, and Korea are not part of the baseline pass-through panels.

The inflation regressions use a more restricted sample, defined according to the comparability of the pass-through mechanism. The baseline emerging-market group includes Brazil, Mexico, Chile, South Africa, Turkey, and India. These are the emerging economies with floating currencies whose daily dollar exchange rates are available over the whole 2006–2026 network sample from the sources used to build the network (the Federal Reserve H.10 release, the Banco Central de Chile, and, for the lira, Yahoo Finance). They combine recurrent exposure to external financial shocks with large and frequent exchange rate movements, and together they span a wide range of inflation levels and monetary conditions (Appendix Table A3). All six operate inflation-targeting frameworks with floating exchange rates for most of the sample, and all six are classified as inflation targeters in the de facto classification of the IMF (**?**), with two qualifications. India formally adopted flexible inflation targeting in 2015–2016, and Turkey, while formally keeping its framework, cut policy rates in 2021–2023 despite rising inflation, and its exchange rate arrangement was classified as crawl-like in 2023. The regression sample runs from 2010M01, the first month of the monthly macroeconomic panel, to 2026M03 (195 months per country). The first four years of the daily network, which include the initialization of the time-varying filter and the 2008–2009 crisis, therefore do not enter the pass-through regressions. The number of observations falls from 1,146 at $h = 0$ to 1,074 at $h = 12$ because the cumulative outcome requires future prices. Descriptive statistics by country are reported in Table A3 in the Appendix. The advanced-economy comparison group includes the euro area, Japan, and Canada, the advanced economies whose currencies are nodes of the main network and that publish monthly consumer prices. Because a comparison that rests on three clusters has little power, we also use an expanded group, AE7, that adds the United Kingdom, Switzerland, Sweden, and Norway. Their currencies, from the H.10 release, enter a 21-node version of the network (N21) that is otherwise identical to the main one; their consumer prices are the IMF headline indices and their policy rates are from the Bank for International Settlements. The euro area is treated as a single advanced-economy unit because the euro is the relevant currency in the exchange rate network and monetary policy is conducted at the area-wide level by

the European Central Bank. Accordingly, inflation and policy variables for the euro area are measured using aggregate euro area indicators rather than individual member-country series. Australia is part of the network but is excluded from the pass-through regressions because its consumer price index is published only at a quarterly frequency.

The United States is excluded from the pass-through panels because the dollar's dominant-currency status makes its pass-through not comparable with that of the other economies; a sensitivity exercise adds it to the advanced group, with the inverse of the broad dollar index as its exchange rate. Korea, whose classification between emerging and advanced is ambiguous over the sample, is used only in sensitivity exercises, and China, which does not combine inflation targeting with a floating exchange rate, is a node of the network but not part of any panel.

To assess whether the results generalize beyond six countries, we also estimate the model on an expanded panel defined by an explicit inclusion rule: emerging economies that operated an inflation-targeting regime with a floating exchange rate for most of the sample, according to the de facto classification of the IMF (**?**), and for which daily exchange rates and monthly consumer prices are available. The rule adds Colombia, Peru, Indonesia, the Philippines, Thailand, Poland, Hungary, and the Czech Republic, for a total of fourteen emerging economies; all eight are classified by the IMF as inflation targeters with floating or free-floating arrangements. The Czech koruna was kept above a floor against the euro between November 2013 and April 2017, so the Czech Republic did not float freely during that period. Malaysia, which does not target inflation, is excluded. Other economies classified by the IMF as inflation targeters with floating arrangements are not included for reasons of classification or data. Korea and Israel are classified by the IMF as advanced economies, and Korea is used in sensitivity exercises; the Russian ruble ceased to be market-determined after the capital controls and sanctions of 2022; and for the remaining economies, which are smaller and have thinly traded currencies, the daily exchange rate series of the Bank for International Settlements, the source used for the other currencies of the expanded network, do not cover the whole 2006–2025 network period. The expanded panel uses a 25-node network (N25) that adds the eight new currencies to the main network; because the daily series of the Bank for International Settlements used for these currencies end in July 2025, the expanded network and panel end in that month.

Table 2 summarizes the networks and panels used in the article and the question each one answers; Appendix Table A4 lists the variables of each empirical block. Throughout the results, networks are referred to by the labels in the first column and panels by EM6 (the six baseline emerging economies), AE3 (the euro area, Japan, and Canada), AE7 (the expanded advanced group), and EM14 (the expanded emerging panel).

The exchange rate variables used in the local projections are monthly logarithmic percentage changes of end-of-month exchange rates, in units of domestic currency per U.S. dollar, so that positive values represent domestic currency depreciation. For the emerging economies, the rates are Yahoo Finance closing quotes; for the advanced economies, the Federal Reserve H.10 rates. Inflation is measured with the headline consumer price index of each country at the monthly frequency, not seasonally adjusted; seasonal patterns common to all countries are absorbed by the month fixed effects, and the lags of monthly inflation capture part of the country-specific dynamics. Policy rates are taken from the IMF Monetary and Financial Statistics, except for India, for which we use the OECD call money rate.

Table 2: Map of the empirical design: networks, panels, and the question each one answers

Label	Nodes	Period	Panels	Question	Results
N17	Main network: 12 currencies, broad dollar, VIX, EPU United States (daily), EPU China and Russia (monthly)	2006–2026	EM6, EM6 + AE3	Baseline estimates	Sections 4.1–4.6
N17-up	Same nodes, good-volatility layer	2006–2026	EM6	Is the association specific to the bad-volatility layer?	Table 5
N15	N17 without the monthly EPU indices	2006–2026	EM6	Does persistence depend on the monthly nodes?	Table 6
N17-P	N17 with daily proxies (VXFXI, RVI) in place of the monthly EPU indices	2013-11 to 2022-02	EM6	Same question, where daily proxies exist	Table 6
N25	N17 plus COP, PEN, IDR, PHP, THB, PLN, HUF, CZK	2006 to 2025-07	EM14, EM6	Does the association generalize?	Table 7
N23	N25 without the monthly EPU indices	2006 to 2025-07	EM14	Generalization without the monthly nodes	Table 6
N21	N17 plus GBP, CHF, SEK, NOK	2006–2026	EM6 + AE7	Does the contrast with advanced economies survive more clusters?	Table 4
N17 variants	$H = 100$ with cuts at $\pi/2$, $\pi/5$, $\pi/22$; euro numeraire	2006–2026	EM6	Frequency location of the information; choice of numeraire	Appendix

Panels

EM6	Brazil, Chile, India, Mexico, South Africa, Turkey; 2010M01–2026M03. EM5 excludes Turkey.
AE3	Euro area, Japan, Canada (comparative panel EM6+AE3); Korea and the United States in sensitivity exercises.
AE7	AE3 plus United Kingdom, Switzerland, Sweden, Norway; 2009M01–2025M07, with the N21 network.
EM14	EM6 plus Colombia, Peru, Indonesia, Philippines, Thailand, Poland, Hungary, Czech Republic; 2010M01–2025M07. EM13 excludes Turkey.

Note: All networks are TVP-VAR frequency-connectedness networks of daily GJR-GARCH bad (or, for N17-up, good) volatilities, with the specification of Section 3.2. BadNet is the monthly average of each currency's received connectedness in the corresponding network.

3.7. Inference

The local projections are estimated by ordinary least squares in panel form, with fixed effects and observable controls. The interpretation of the coefficients does not rely on the assumption of homoskedastic, independent, and normally distributed errors. In macroeconomic panels, residuals are expected to display cross-country heterogeneity, temporal persistence, within-country dependence, and exposure to common shocks. Local projection residuals are, in addition, serially correlated by construction at horizons $h > 0$.

Inference with six clusters requires particular care, because cluster-robust standard errors are biased downward and conventional critical values over-reject when the number of clusters is small (Cameron et al., 2008; MacKinnon and Webb, 2017). We therefore report several procedures side by side rather than relying on a single one. First, standard errors clustered by country (CR1) with critical values from a t distribution with $G - 1$

degrees of freedom. Second, a wild cluster bootstrap with the null hypothesis imposed and the six-point weights of Webb (2023), which remain informative with very few clusters; the baseline uses 9,999 replications. Third, Driscoll and Kraay (1998) standard errors, which are robust to cross-sectional dependence and serial correlation. Fourth, the Ibragimov and Müller (2010) test, which estimates the model separately for each country and applies a t test with $G-1$ degrees of freedom to the country-level coefficients; this test is valid under heterogeneous coefficients but has low power with six countries. Fifth, a permutation test that reassigns BadNet series across countries. Because the model is estimated at thirteen horizons, we also report a sup-t simultaneous band (Montiel Olea and Plagborg-Møller, 2019) and p-values adjusted by the Holm (1979) and Benjamini and Hochberg (1995) procedures. A Monte Carlo experiment calibrated to the panel, reported in Section 4.6.1, documents the size of each procedure in this design.

Because BadNet is a generated regressor constructed in previous stages, we also implement a two-step bootstrap that re-estimates the network measure in each replication. Re-estimating the TVP-VAR in every replication is computationally prohibitive, so the first step is approximated by a rolling-window VAR(2) estimated on the 200 trading days ending on the last day of each month, with the same forecast horizon and generalized decomposition. In each of 199 replications, the daily VAR residuals in every window are multiplied by Rademacher weights, the VAR and the decomposition are re-estimated, monthly exposure is recomputed, and the local projection is re-estimated with a wild cluster bootstrap in the second step. The ratio between the two-step and the second-step standard deviations of the interaction coefficient measures the additional uncertainty due to the generated regressor.

3.8. Interpretation, Identification, and Limits of the Strategy

The interpretation of the results should be understood as evidence that ERPT is associated with the state of the bad-volatility currency network, rather than as a structural causal identification of the transmission channels. Neither depreciation nor BadNet is exogenous: both respond to domestic and global shocks, and the design does not isolate exogenous exchange rate movements. The coefficients are therefore conditional associations. The empirical strategy seeks to mitigate omitted-variable concerns through four elements: country fixed effects, month fixed effects, lagged dynamics, and a set of identification checks.

Country fixed effects absorb structural characteristics that change slowly, such as trade openness, exchange rate regime, financial depth, monetary credibility, productive structure, and the composition of the consumption basket. Month fixed effects absorb shocks common to the countries in the sample, such as global crises, changes in the international financial cycle, U.S. monetary policy, the global dollar, and simultaneous commodity shocks. As a result, the coefficient on the interaction is identified only from differences across countries, in the same month, in the exposure of each currency to the bad-volatility network. The lags of inflation, depreciation, and BadNet control for the predictable component of each variable.

This structure is essential because BadNet could otherwise be confounded with conventional indicators of global stress. The conceptual difference is that the VIX, EPU, and international interest rates measure observable global states, whereas BadNet measures the directional propagation of bad-volatility shocks through the currency network. Therefore, the empirical contribution of BadNet should be interpreted as the additional information contained in the currency's position within the bad-volatility network, beyond the common

global shocks absorbed by the month fixed effects. Section 4.9 shows that BadNet is only partly explained by the VIX, the broad dollar, and the aggregate connectedness index.

Four exercises address the remaining identification concerns. First, BadNet lagged by one month replaces its contemporaneous value, reducing concerns that the interaction captures immediate feedback from domestic depreciation or inflation to the network measure. Second, pre-trend placebos regress past cumulative inflation, $100[\ln CPI_{i,t-1} - \ln CPI_{i,t-1-k}]$, on the same regressors. Third, the sensitivity analysis of Oster (2019) reports how strong selection on unobservables would have to be, relative to selection on the observed controls, to drive the interaction to zero. Fourth, an instrumental-variable version of the local projection (Ramey and Zubairy, 2018) instruments depreciation and its interaction with BadNet by the U.S. monetary policy surprises of Jarociński and Karadi (2020) and by the product of the surprise and lagged BadNet. Since these surprises can also affect emerging-market inflation through commodity prices and global demand, the exclusion restriction is questionable, and we treat this exercise as a check on the sign rather than as an identification strategy. Because these surprises are common to all countries, month fixed effects cannot be included; the instrumental-variable specification uses country fixed effects and observable global controls (the VIX and the monthly changes in oil prices, the broad dollar, global food prices, the ten-year Treasury yield, and the federal funds rate).

Robustness analyses vary the monthly aggregation of BadNet, the network measure (aggregate connectedness, the exposure of the other emerging currencies, the connectedness the currency transmits and its net position, and networks of total volatility), the volatility layer, the frequency band, the lag structure, the treatment of extreme observations, the numeraire, the composition of the panel, and the composition of the network (Table 2). Together, they assess whether the estimated pattern reflects directional exposure to the bad-volatility network, generic exchange rate connectedness, common global shocks, contemporaneous feedback, extreme observations, or the specific composition of the sample.

4. Results

This section presents the results. Sections 4.1 to 4.3 establish the main association: in emerging economies, but not in advanced ones, pass-through is lower when BadNet is high. Sections 4.4 and 4.5 examine the frequency of the relevant spillovers and the role of monetary policy. Sections 4.6 to 4.9 ask how robust and general the association is, with respect to inference with few clusters and alternative specifications, the composition of the network, the country sample, and identification checks. Section 4.10 confronts the results with the remaining channels of Section 2.4, and Section 4.11 brings the evidence together. Throughout, the baseline is the N17 network with the EM6 panel, and supporting tables are in the Appendix.

4.1. BadNet and Pass-Through in Emerging Economies

Under the amplification hypothesis of Section 2.4, depreciations that occur when the domestic currency is more exposed to bad-volatility spillovers would carry stronger cost pressures and a higher risk of unanchoring expectations, and the interaction would be positive. The results point in the opposite direction. Table 3 reports the coefficient of the interaction between currency depreciation and BadNet across different horizons, together with the pass-through of depreciation at the mean of BadNet.

The interaction coefficient is negative at all horizons. This means that, for emerging economies, exposure to BadNet is associated with a lower marginal slope of the relationship

Table 3: Emerging-economy panel: interaction between currency depreciation and BadNet

	$h0$	$h1$	$h3$	$h6$	$h12$
$\Delta e^{+}_{i,t} \times BadNet_{i,t}$	−0.0065	−0.0752***	−0.1522**	−0.1684**	−0.1873**
	(0.0045)	(0.0035)	(0.0194)	(0.0230)	(0.0189)
p-value, CR1, $t(G-1)$	0.204	< 0.001	0.001	0.001	< 0.001
p-value, wild cluster bootstrap	0.133	0.003	0.012	0.011	0.011
p-value, Driscoll–Kraay	0.328	< 0.001	0.005	0.035	0.069
p-value, Ibragimov–Müller	0.540	0.150	0.132	0.249	0.219
$\Delta e^{+}_{i,t}$ (at mean BadNet)	0.0055	0.0686**	0.1775**	0.2467**	0.3886**
	(0.0040)	(0.0223)	(0.0464)	(0.0647)	(0.1135)
Observations	1,146	1,140	1,128	1,110	1,074
R^2	0.699	0.712	0.711	0.695	0.735

Note: The dependent variable is $100[\ln CPI_{i,t+h} - \ln CPI_{i,t-1}]$. BadNet is standardized on the emerging panel, so the interaction is the change in the cumulative pass-through of a 1% depreciation, in percentage points, per standard deviation of BadNet. All regressions include country and month fixed effects and three lags of monthly inflation, depreciation, BadNet, and their interaction. Standard errors clustered at the country level (CR1) are reported in parentheses. The Ibragimov–Müller test applies a $t(5)$ test to the six country-level coefficients; all inference procedures are compared in Appendix Table A5. Stars refer to the wild cluster bootstrap p-value for the interaction (Webb weights, 9,999 replications, null imposed) and to the CR1 p-value with $t(G-1)$ critical values for the main effect. Because the bootstrap p-values are larger than those implied by CR1 standard errors, stars in this and later tables are fewer than the ratio of coefficient to standard error would suggest. The emerging-economy panel includes Brazil, Mexico, Chile, South Africa, Turkey, and India, 2010M01–2026M03. Significance: $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

between currency depreciation and inflation. The magnitude of the effect increases over the horizon, from −0.0065 at $h0$, which is not statistically significant, to −0.075 at $h1$ and −0.187 at $h12$. The magnitude is economically large. At $h6$, the cumulative pass-through of a 1% depreciation at the mean of BadNet is 0.25 percentage points; a one-standard-deviation increase in BadNet is associated with a reduction of 0.17 points, about two-thirds of that pass-through. Moving from the 25th to the 75th percentile of BadNet reduces the six-month pass-through from 0.35 to 0.13 points (−62%) and the twelve-month pass-through from 0.50 to 0.26 points (−48%). Figure 1 shows the full path of the interaction from $h0$ to $h12$, with pointwise and simultaneous bands. The gradual profile is consistent with the dynamics of ERPT documented in the literature, since contracts, intermediate input costs, pricing decisions, expectations, and monetary policy responses do not adjust fully at the moment of depreciation (Campa and Goldberg, 2005; Burstein and Gopinath, 2014).

BadNet should therefore not be read as a multiplier of inflationary shocks. It identifies the composition of the depreciation. When received connectedness is high, the domestic currency is moving mostly with the rest of the network rather than in response to its own shocks. Section 2.4 lists the channels through which such a depreciation could pass through less, and Section 4.10 examines which of them find support in the data.

4.2. Heterogeneity between Emerging and Advanced Economies

The second block of results expands the analysis to a comparative panel of emerging and advanced economies. The purpose is to separate two dimensions that are jointly observed in the emerging-economy panel: the average effect of currency depreciation on inflation and the conditioning effect of BadNet on the marginal slope of pass-through. To do so, we estimate a triple interaction among currency depreciation, BadNet, and an indicator for emerging economies. The advanced group includes the euro area, Japan, and

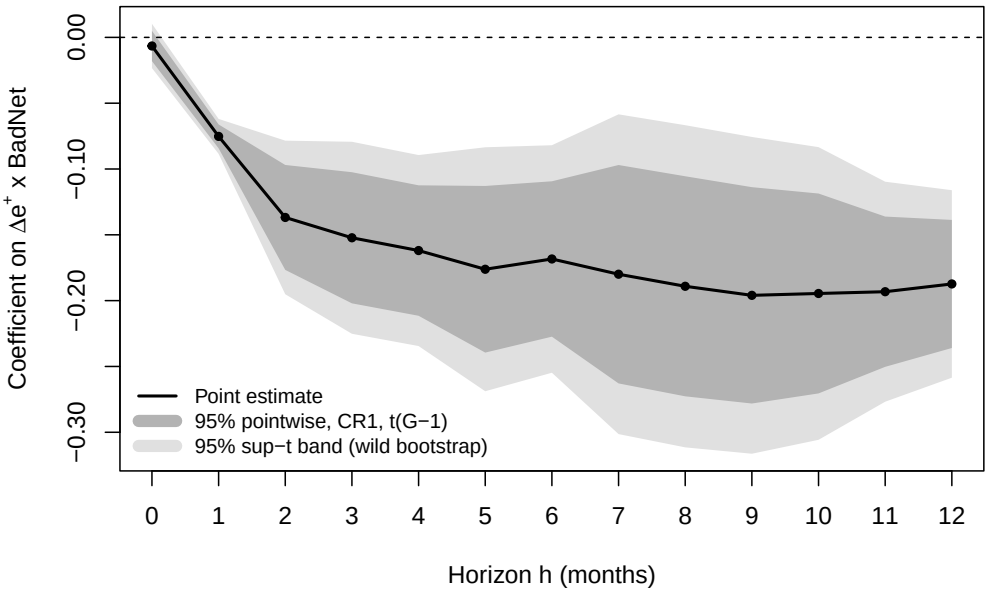


Figure 1: Interaction between depreciation and BadNet, horizons 0 to 12

Note: Coefficient on $\Delta e^+_{i,t} \times BadNet_{i,t}$ in the baseline specification, emerging-economy panel. The dark band is the pointwise 95% interval based on CR1 standard errors and $t(G-1)$ critical values; the light band is the 95% sup-t simultaneous band across horizons.

Canada, so the comparative panel has nine clusters.

Table 4 reports the key coefficients from the comparative specification.

Table 4 reveals three main results. First, pass-through in advanced economies is close to zero at all horizons, between -0.02 and 0.03 points per 1% depreciation, in contrast with 0.07 to 0.39 points in the emerging panel (Table 3). This confirms that average ERPT is much stronger in emerging economies. Second, the interaction $\Delta e^+ \times BadNet$, which captures the conditioning effect of BadNet in the reference group of advanced economies, is small, positive, and statistically insignificant at all horizons. This indicates that BadNet does not operate as a universal moderator of exchange rate pass-through.

Third, the triple interaction $\Delta e^+ \times BadNet \times Emerging$ is negative at all horizons, between -0.10 and -0.38 from $h1$ onward. Its precision depends on the inference procedure. It is significant from $h1$ onward with CR1 standard errors and $t(G-1)$ critical values and with Driscoll–Kraay standard errors, but not with the wild cluster bootstrap, whose p-values lie between 0.18 and 0.28. This is expected in a design in which the contrast rests on three advanced clusters (MacKinnon and Webb, 2017). Table 4 therefore complements the regression in two ways. The first is the Ibragimov and Müller (2010) comparison of country-level coefficients, which treats each economy as one observation. The second is the expanded advanced group, AE7, which more than doubles the number of advanced clusters.

Because the triple interaction is a relative effect, the substantive interpretation for emerging economies requires computing the total effect of BadNet, defined as the sum of $\Delta e^+ \times BadNet$ and $\Delta e^+ \times BadNet \times Emerging$. The middle part of Table 4 reports this total marginal effect.

The total effect of BadNet on pass-through in emerging economies is negative and statistically significant from $h1$ onward. Its magnitude increases in absolute value from -0.079 at $h1$ to -0.212 at $h12$ and is close to the baseline estimate of Table 3, indicating that adding the advanced economies does not change the emerging-market result. The group-level comparison confirms the contrast: the average country-level interaction is

Table 4: Comparative model and total effect of BadNet in emerging economies

	$h0$	$h1$	$h3$	$h6$	$h12$
Δe^+ (advanced, BadNet at EM mean)	−0.0178	−0.0081	−0.0184	0.0224	0.0327
	(0.0087)	(0.0149)	(0.0220)	(0.0245)	(0.0388)
$\Delta e^+ \times BadNet$ (advanced, β_3)	0.0080	0.0200	0.0681	0.0779	0.1706
	(0.0139)	(0.0228)	(0.0455)	(0.0697)	(0.1027)
p-value, CR1, $t(G-1)$	0.580	0.405	0.173	0.296	0.135
$\Delta e^+ \times BadNet \times Emerging$ (β_6)	−0.0150	−0.0988	−0.2263	−0.2569	−0.3828
	(0.0129)	(0.0270)	(0.0643)	(0.0963)	(0.1295)
p-value, CR1, $t(G-1)$	0.278	0.006	0.008	0.028	0.018
p-value, wild cluster bootstrap	0.277	0.178	0.175	0.242	0.210
p-value, Driscoll–Kraay	0.388	0.001	< 0.001	0.003	< 0.001
Total effect ($\beta_3 + \beta_6$)	−0.0070	−0.0788***	−0.1582***	−0.1790***	−0.2122***
	(0.0038)	(0.0075)	(0.0236)	(0.0299)	(0.0332)
p-value, CR1, $t(G-1)$	0.100	< 0.001	< 0.001	< 0.001	< 0.001
Ibragimov–Müller comparison of country-level coefficients					
Mean, emerging ($G = 6$)	0.002	−0.023	−0.076	−0.069	−0.091
Mean, advanced ($G = 3$)	0.036	0.042	0.075	0.077	0.114
p-value of the difference	0.159	0.027	0.017	0.041	0.026
Sensitivity of β_6 to the advanced group					
Korea as advanced ($G = 10$)	−0.016	−0.095***	−0.226***	−0.245**	−0.370**
	(0.011)	(0.024)	(0.060)	(0.088)	(0.116)
United States as advanced ($G = 10$)	−0.016	−0.101***	−0.220***	−0.254**	−0.362***
	(0.012)	(0.021)	(0.051)	(0.083)	(0.105)
Seven advanced economies (AE7), N21 network ($G = 13$)					
Δe^+ (advanced, BadNet at EM mean)	−0.010	0.005	0.009	0.043	0.089
	(0.008)	(0.009)	(0.018)	(0.018)	(0.046)
$\Delta e^+ \times BadNet$ (advanced, β_3)	0.016	0.028*	0.045*	0.033	0.065
	(0.011)	(0.012)	(0.025)	(0.026)	(0.043)
$\Delta e^+ \times BadNet \times Emerging$ (β_6)	−0.024**	−0.110**	−0.213**	−0.228*	−0.285**
	(0.010)	(0.018)	(0.046)	(0.053)	(0.065)
Total effect ($\beta_3 + \beta_6$)	−0.007*	−0.082***	−0.168***	−0.195***	−0.220***
	(0.004)	(0.008)	(0.024)	(0.031)	(0.033)
Ibragimov–Müller difference, p	0.123	0.019	0.061	0.123	0.029
Observations	1,738	1,729	1,711	1,684	1,630
R^2	0.654	0.686	0.706	0.704	0.750

Note: Comparative panel of six emerging and three advanced economies (Euro area, Japan, Canada), $G = 9$. The specification includes Δe^+, BadNet, their interaction, their interactions with the emerging-economy indicator, country and month fixed effects, and the baseline lags. The advanced-economy group is the reference category. Standard errors clustered at the country level are reported in parentheses. With only three advanced clusters, the wild cluster bootstrap can be severely conservative for coefficients that contrast groups (MacKinnon and Webb, 2017), unlike in the baseline design, where it over-rejects (Appendix Table A5); the Ibragimov–Müller rows report a group-level test that does not rely on clustered standard errors. The total effect is $\beta_3 + \beta_6$. The Ibragimov–Müller rows estimate the baseline interaction separately for each country and compare the group means with a two-sample t test. Stars on the total effect and on the Korea and United States rows refer to CR1 p-values with $t(G-1)$ critical values. In the AE7 block, BadNet comes from the N21 network and is standardized with the emerging-economy moments of that network; stars on β_3 and β_6 refer to wild cluster bootstrap p-values (1,999 replications), on the total effect to CR1 p-values with $t(G-1)$ critical values. Significance: $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

negative in emerging economies and positive in advanced economies, and the difference is significant at the 5% level from $h1$ to $h12$. The result is also unchanged when Korea or the United States is added to the advanced group.

The expanded advanced group confirms the contrast with more power. With the United Kingdom, Switzerland, Sweden, and Norway added, and BadNet measured in the N21 network, which includes their currencies, pass-through in the seven advanced economies remains small, between -0.01 and 0.09 points per 1% depreciation, and the interaction with BadNet in advanced economies is small and significant only at the 10% level at $h1$ and $h3$. The triple interaction now rejects with the wild cluster bootstrap, at the 5% level at $h0$, $h1$, $h3$, and $h12$ and at the 10% level at $h6$, and the total effect for emerging economies is virtually identical to that of the baseline comparison (-0.195 at $h6$). The group-level comparison rejects at $h1$ and $h12$ at the 5% level and at $h3$ at the 10% level, but not at $h6$. Re-estimating the comparison with the three original advanced economies and the N21 network gives values close to those of the main network (a triple interaction of -0.249 at $h6$), so the gain in precision comes from the additional clusters rather than from the change of network.

The negative total effect does not imply that emerging economies are less vulnerable to exchange rate movements. Rather, it shows that their higher average ERPT depends on the network state in which depreciation occurs. The absence of the effect in advanced economies is consistent with the channels of Section 2.4: their pass-through is close to zero at every level of BadNet, so there is little pass-through for the network state to modify.

4.3. Marginal Effects across the Distribution of BadNet

The substantive interpretation of models with interactions should not rely only on the direct reading of coefficients. For this reason, Figure 2 presents the marginal effect of currency depreciation on inflation across the distribution of BadNet in the emerging-economy panel, at horizons of one, six, and twelve months. Because pass-through in advanced economies is close to zero at every level of BadNet (Table 4), the figure focuses on the emerging panel.

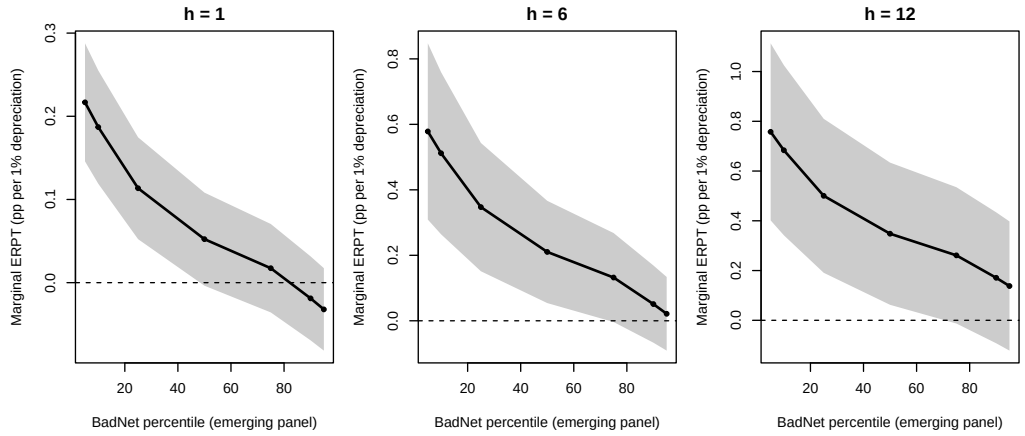


Figure 2: Marginal ERPT across BadNet percentiles, emerging economies

Note: Cumulative pass-through of a 1% depreciation, $\beta_1 + \beta_3 BadNet$, evaluated at percentiles of the empirical distribution of BadNet in the emerging panel, with 95% intervals based on CR1 standard errors and $t(G-1)$ critical values. Baseline specification.

Figure 2 translates the interaction terms into an economically readable object: the marginal slope of ERPT at different levels of the domestic currency's exposure to BadNet. Substantively, it asks whether a currency depreciation of the same magnitude generates

the same inflationary effect when the currency is weakly or strongly exposed to the global network of bad-volatility transmission.

At low percentiles of BadNet, the marginal effect of depreciation on inflation is positive and large, reflecting the stronger inflation sensitivity to exchange rate movements documented for emerging economies (Choudhri and Hakura, 2006; Ha et al., 2020). At the 10th percentile, the cumulative six-month pass-through is about 0.5 points per 1% depreciation. As exposure to BadNet increases, however, the marginal effect declines. At six months, pass-through is no longer statistically different from zero from the 75th percentile onward. Within the range shown, which covers 90% of the observed distribution, the point estimate remains positive at every horizon; it would cross zero only near the 97th percentile at six months, at the edge of the empirical support, and it is never significantly negative. This downward slope indicates that the higher average ERPT of emerging economies is not constant across network states, but depends on the degree to which the domestic currency is exposed to bad-volatility spillovers.

This result helps reconcile two findings that might otherwise appear contradictory. On the one hand, emerging economies exhibit stronger average ERPT. On the other hand, the interaction among depreciation, BadNet, and the emerging-economy indicator is negative. Figure 2 shows that these results are compatible: emerging economies start from a higher inflation sensitivity to depreciation, but this sensitivity declines when depreciation occurs under greater exposure to BadNet. Pass-through is lower, but not reversed, over the supported range of BadNet.

4.4. Frequency Decomposition: Short-Run and Persistent BadNet

If BadNet captured mainly high-frequency market noise, its effect on pass-through should come from short-run spillovers. Appendix Table A6 reports the interaction of depreciation with the short-run and persistent components of BadNet, each standardized, for the baseline network ($H = 10$) and for networks estimated with $H = 100$ and three cut-off frequencies.

In the baseline network, the persistent component reproduces the aggregate result: its interaction with depreciation is negative and significant from $h1$ to $h12$, with magnitudes very close to those of Table 3. The short-run component also has negative point estimates at all horizons, of similar size, but they are imprecise and significant only at the 10% level at $h3$. The conditioning effect is therefore not driven by the highest-frequency spillovers, which would be the case if BadNet captured mainly transitory market noise.

With a longer forecast horizon, which yields a finer frequency grid, the aggregate result is unchanged (-0.156 at $h6$), and the location of the information depends on the cut-off. With a cut at two days, it lies in the persistent band; with a cut at 22 trading days, it lies in the higher-frequency band, which strictly contains cycles of up to about two months, and the slowest component loses precision. Taken together, the evidence indicates that the relevant information is in spillovers that last more than a few trading days but no more than one to two months. These are episodes of sustained, but not permanent, bad-volatility transmission across currencies.

4.5. High-Interest-Rate Regimes and Macro-Financial Interpretation

The defensive policy channel of Section 2.4 predicts that central banks raise rates more after high-BadNet depreciations. Appendix Table A7 tests this directly, and also asks whether the dampening differs when depreciation occurs in a high-interest-rate regime. Because policy-rate levels differ widely across countries, the regime is defined within each country.

Defining the regime within each country matters, because a pooled upper-quartile threshold would classify mostly the observations of a few high-rate economies as high-rate states. With the within-country median, the triple interaction is negative and significant from $h1$ onward, reaching -0.33 at $h12$; the continuous version is negative and significant at $h6$ and $h12$. The upper-quartile indicator, which leaves about 25% of each country's months in the high regime, is negative at $h6$ and $h12$ but imprecise. The dampening associated with BadNet is therefore stronger when depreciation occurs while the domestic policy rate is already high relative to the country's own history.

The last row of Appendix Table A7 shows that this is not because central banks respond more aggressively to depreciation when BadNet is high. When the dependent variable is the cumulative change in the policy rate, the interaction is negative: in high-BadNet states, depreciations are followed by smaller rate increases over the first six months. The defensive policy channel is therefore rejected: the lower pass-through does not reflect a stronger monetary response.

There is, however, an important limitation. This specification should be interpreted as regime evidence, not as causal identification of a monetary policy mechanism. High-interest-rate regimes may reflect both policy responses and underlying macroeconomic conditions, such as persistent inflation, unanchored expectations, external shocks, or exchange rate instability. Section 4.10 shows that an analogous pattern appears when the regime is defined by inflation above the country median, which suggests that the relevant state is a broader macro-financial one.

4.6. Robustness and Stability of the Empirical Pattern

The robustness analyses examine whether the negative association between BadNet and pass-through in emerging economies survives alternative inference procedures, specifications, and sample definitions. We report all the exercises we conducted, including those that weaken the result.

4.6.1. Inference with few clusters

The first dimension of robustness concerns inference. Appendix Table A5 reports, for the baseline interaction, the p-values of all the procedures described in Section 3.7, the sup-t simultaneous band, the multiple-testing adjustments, the permutation test, the two-step bootstrap that accounts for the generated regressor, and the size of each test in a Monte Carlo experiment calibrated to the panel.

The results support a nuanced reading of the evidence. The interaction is significant from $h1$ to $h12$ with the wild cluster bootstrap and with Driscoll–Kraay standard errors (at the 10% level at $h12$), the sup-t band excludes zero jointly from $h1$ to $h12$, and the false discovery rate adjustment leaves the conclusion unchanged. The Holm adjustment, which is more conservative, keeps $h1$ significant at 5%. The two-step bootstrap shows that the first-stage uncertainty roughly doubles the standard error, but the 95% interval still excludes zero at $h6$ and $h12$. This exercise bounds rather than measures the generated-regressor problem, because its first step replaces the TVP-VAR by a rolling-window VAR whose exposure measure has a correlation of only 0.52 with the baseline BadNet, and its point estimates (-0.208 and -0.243) differ accordingly from the baseline.

Two procedures are less favorable. The Ibragimov and Müller (2010) test does not reject at conventional levels, because one of the six country-level coefficients (Chile) is positive and the others are heterogeneous in size; the permutation test gives p-values of 0.09 and 0.13 at $h6$ and $h12$. The Monte Carlo experiment helps to weigh these results. Under the null, the CR1 test with $t(G-1)$ critical values rejects in 19% of the replications

and the wild cluster bootstrap in 13%, while the Ibragimov–Müller test is close to its nominal size. With six clusters, therefore, none of the cluster-robust procedures is exact, and the Ibragimov–Müller test, which is correctly sized, has little power. We read the six-country evidence as suggestive, and the expanded panel of Section 4.8 provides the main answer to this limitation, since with fourteen countries both the group-level test and the permutation test reject.

4.6.2. Alternative specifications

Table 5 reports the interaction coefficient under the main alternative specifications, and Appendix Table A8 reports the complete set. Several features stand out. First, lagged BadNet produces results close to the baseline (-0.130 at $h6$ and -0.161 at $h12$), which reduces concerns that the contemporaneous interaction reflects immediate feedback from domestic depreciation to the network measure. Second, alternative monthly aggregations (maximum and end-of-month), year-on-year inflation at $t+h$ as the outcome, twelve lags, a balanced sample, and the replacement of month fixed effects by observable global controls all preserve the sign and significance of the result. Third, the result is not absorbed by interactions of depreciation with the VIX, the broad dollar, U.S. policy uncertainty, a risk-off indicator, or the lagged policy rate. Once BadNet is included, the interactions of depreciation with these observable indicators are small and insignificant, except for the VIX interaction at $h1$.

Three results in Table 5 limit the interpretation and deserve emphasis. First, the effect is specific to the currency's own exposure. Neither the aggregate connectedness index nor the average exposure of the other emerging currencies, which capture a common network state, reproduces the result. This supports the interpretation of BadNet as a currency-specific position in the network rather than a proxy for global stress, which is in any case absorbed by the month fixed effects. Appendix Table A9 adds the other currency-specific measures. The connectedness the currency transmits gives results close to the baseline, while its net position is weaker and significant only at short horizons, so what matters is how much the currency moves together with the rest of the network, in either direction, rather than whether it is a net transmitter or receiver. Second, the effect is specific to the bad-volatility layer. In networks of total volatility, received connectedness gives a weaker result when it is built from the GJR-GARCH variance, which still carries the asymmetric response to depreciation, and no result when it is built from a symmetric sGARCH variance (Appendix Table A9). The good-volatility network, built in the same way from appreciation-day volatility, has a smaller negative association at $h1$ and none beyond it, and when both layers enter jointly only the bad-volatility interaction is negative. Third, the result depends on the dollar as numeraire. When all exchange rates are expressed against the euro, the interaction is significant only at $h1$. Since most emerging-market trade and financing is denominated in dollars (Gopinath et al., 2020; Boz et al., 2022), the network relevant for domestic prices is plausibly the one measured against the dollar, but this result shows that BadNet is not invariant to the choice of reference currency.

Two further checks address the possibility that the interaction captures a nonlinearity in the size of depreciations rather than the network state. Because received connectedness falls when the currency's own volatility rises, large depreciations tend to coincide with lower BadNet; if large depreciations passed through proportionally less, the interaction could reflect that convexity. Table 5 shows that controlling for squared depreciation leaves the interaction negative and significant at all horizons (-0.145 at $h6$), and Appendix Table A8 shows that interacting depreciation with the currency's own past volatility leaves

Table 5: Robustness of the interaction between depreciation and BadNet, main specifications

Specification	$h1$	$h3$	$h6$	$h12$
Baseline (N17)	-0.075^{***}	-0.152^{**}	-0.168^{**}	-0.187^{**}
	(0.004)	(0.019)	(0.023)	(0.019)
BadNet lagged $(t-1)$	-0.072^{***}	-0.139^{**}	-0.130^{***}	-0.161^{***}
	(0.004)	(0.017)	(0.010)	(0.017)
Year-on-year inflation at $t+h$	-0.091^{**}	-0.199^{**}	-0.213^{**}	-0.178^{**}
	(0.023)	(0.050)	(0.039)	(0.020)
Twelve lags	-0.084^{***}	-0.171^{***}	-0.195^{**}	-0.191^{**}
	(0.010)	(0.026)	(0.040)	(0.033)
Country FE and global controls, no month FE	-0.079^{***}	-0.170^{**}	-0.207^{*}	-0.274^{**}
	(0.009)	(0.030)	(0.041)	(0.045)
+ squared depreciation, $(\Delta e^{+})^2$	-0.052^{**}	-0.138^{***}	-0.145^{***}	-0.154^{***}
	(0.011)	(0.025)	(0.023)	(0.017)
+ lagged policy rate	-0.079^{***}	-0.158^{***}	-0.176^{**}	-0.193^{***}
	(0.005)	(0.021)	(0.026)	(0.022)
+ $\Delta e^{+}\times$ VIX and $\Delta e^{+}\times$ dollar	-0.063^{***}	-0.142^{***}	-0.162^{***}	-0.183^{***}
	(0.006)	(0.017)	(0.021)	(0.018)
TCI^{B} (global state)	-0.024^{*}	-0.064	-0.041	-0.097
	(0.013)	(0.038)	(0.047)	(0.090)
External state (mean of other EMs)	0.013^{*}	-0.029	-0.014	-0.062
	(0.008)	(0.060)	(0.094)	(0.123)
Euro numeraire	-0.039^{**}	-0.016	0.022	0.091
	(0.019)	(0.041)	(0.045)	(0.076)
Placebo: good-volatility network (N17-up)	-0.047^{**}	-0.021	0.024	0.057
	(0.016)	(0.058)	(0.090)	(0.116)
Both layers: bad-volatility coefficient	-0.105^{**}	-0.315^{**}	-0.410	-0.486^{**}
	(0.017)	(0.075)	(0.103)	(0.099)

Note: Coefficient on the interaction between depreciation and the indicated network measure, EM6 panel. Standard errors clustered by country (CR1) in parentheses; stars refer to wild cluster bootstrap p-values (Webb weights, null imposed). Unless stated otherwise, regressions follow the baseline specification of Table 3. N17 is the main network and N17-up its good-volatility layer, built from the same nodes (Table 2). "Both layers" includes the bad-volatility and good-volatility interactions in the same regression. The complete set of specifications is in Appendix Table A8. Significance: $^{***}p<0.01$; $^{**}p<0.05$; $^{*}p<0.10$.

it essentially unchanged. When the interaction of squared depreciation with BadNet is also included (Appendix Table A8), the linear interaction is no longer significant and the quadratic one is negative and significant from $h1$ to $h12$ (-0.010 at $h6$, wild bootstrap $p=0.015$): the lower pass-through under high BadNet is concentrated in the larger depreciations. Finally, monthly inflation at $t+h$ (Appendix Table A8) shows that the cumulative effect builds up mainly in the first three months, and a panel VAR(12) in depreciation and inflation estimated separately for months with BadNet below and above its mean gives a cumulative price response to a 1% depreciation of 0.47 percentage points after six months and 0.59 after twelve in the low-BadNet regime, against values between -0.02 and 0.00 in the high-BadNet regime, in line with the local projections.

4.7. Network Composition and the Monthly Uncertainty Nodes

As discussed in Section 3.2, two nodes of the network are monthly uncertainty indices repeated across business days. Table 6 shows how the main result changes with the composition of the network. The comparison separates the role of these nodes from the role of the sample period, because the daily proxies for China and Russia are available only between November 2013 and February 2022.

Three conclusions follow. First, the association at $h1$ appears in every network (N17, N21, N15, N17-P, N25, and N23), and adding four advanced-economy currencies

Table 6: Sensitivity of the interaction to network composition, emerging economies

Network	$h1$	$h3$	$h6$	$h12$	Good vol., $h6$
Full sample (2010–2026), EM6					
N17, main network	-0.075^{***}	-0.152^{**}	-0.168^{**}	-0.187^{**}	0.024
	(0.004)	(0.019)	(0.023)	(0.019)	(0.090)
N21, adding GBP, CHF, SEK, NOK	-0.078^{***}	-0.157^{***}	-0.182^{***}	-0.191^{***}	
	(0.006)	(0.019)	(0.022)	(0.024)	
N15, without monthly EPU indices	-0.037^{***}	-0.050^{*}	-0.044	0.003	0.021
	(0.006)	(0.026)	(0.034)	(0.058)	(0.080)
bad volatility controlling for good	-0.046	-0.074^{***}	-0.104^{**}	-0.057^{*}	
	(0.010)	(0.010)	(0.019)	(0.039)	
Common period of the daily proxies (2013-11 to 2022-02), EM6					
N17	-0.103^{***}	-0.198^{***}	-0.218^{***}	-0.188^{***}	
	(0.010)	(0.028)	(0.023)	(0.022)	
N15	-0.101^{**}	-0.209^{**}	-0.239^{**}	-0.213^{***}	
	(0.011)	(0.035)	(0.039)	(0.023)	
N17-P, daily proxies VXFXI and RVI	-0.100^{**}	-0.196^{**}	-0.222^{**}	-0.194^{**}	-0.220^{**}
	(0.012)	(0.035)	(0.036)	(0.026)	(0.030)
bad volatility controlling for good	-0.114^{**}	-0.241	-0.274	-0.091	
	(0.030)	(0.113)	(0.128)	(0.141)	
EM14 (2010–2025)					
N25, adding eight currencies	-0.057^{***}	-0.116^{***}	-0.127^{***}	-0.148^{***}	
	(0.009)	(0.014)	(0.018)	(0.022)	
N23, without monthly EPU indices	-0.032^{***}	-0.021	0.003	0.045	
	(0.007)	(0.027)	(0.036)	(0.058)	

Note: Coefficient on $\Delta e^{+} \times BadNet$ in the baseline specification. Standard errors clustered by country in parentheses; stars refer to wild cluster bootstrap p-values. Networks are defined in Table 2. N17 is the main network, with twelve currencies, the broad dollar index, the VIX, the daily EPU index for the United States, and the monthly EPU indices for China and Russia. N21 adds the British pound, the Swiss franc, the Swedish krona, and the Norwegian krone. N15 removes the two monthly EPU indices. N17-P replaces them with daily market-based proxies, VXFXI (the CBOE China ETF volatility index) and RVI (the Russian volatility index of the Moscow Exchange). N25 adds the currencies of the eight additional economies of the EM14 panel (Colombia, Peru, Indonesia, the Philippines, Thailand, Poland, Hungary, and the Czech Republic), and N23 is N25 without the monthly EPU indices. The last column is the placebo with the good-volatility network built from the same nodes. In the full-sample N15 network the bad-volatility layer of the monthly nodes is no longer present; in the proxy network the shares of days of depreciation pressure of the two proxies are 0.49 and 0.51, against 0.91 and 0.83 for the monthly indices. Significance: $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

(N21) leaves the baseline essentially unchanged at all horizons. Second, in the common period of the proxies, the result does not depend on how China and Russia enter the network. With the monthly indices, without them, or with daily market-based proxies, the interaction at $h6$ lies between -0.22 and -0.24 and is significant at all horizons. Third, over the full sample, the persistence of the effect beyond the first months depends on the monthly nodes. Without them, the interaction is significant at $h1$ and marginally at $h3$, and the bad-volatility interaction becomes significant at $h3$ and $h6$ only once the good-volatility layer is controlled for. In the fourteen-country panel without the monthly nodes, only $h1$ remains significant. Because no daily proxy exists outside 2013–2022, we cannot determine whether this difference reflects information that the monthly indices carry about China and Russia or the mechanical structure of a step-function node.

Table 6 also shows that the specificity of the bad-volatility layer is not uniform. In the proxy network, the good-volatility placebo is as strong as the bad-volatility measure, and when both enter jointly the bad-volatility layer dominates only at $h1$. In this subperiod, therefore, what matters is exposure to the currency network as such rather than to its

bad-volatility layer specifically. We report these results in the body of the article because they bound the main claim. The association between BadNet and lower pass-through in the first month after a depreciation is robust to network composition; the claim that it persists for a year, and that it is specific to the bad-volatility layer, holds in the main network and in the common period with the original nodes, but not in every alternative network.

4.8. Country Heterogeneity and Generalization

The pooled coefficients of Table 3 average over economies that differ widely in the size and frequency of their depreciations and in their exposure to the network (Appendix Table A3). Estimated country by country (Appendix Table A10), the interaction is close to zero in India, Mexico, and South Africa, positive in Chile, and largest in absolute value in the economies with the largest and most frequent depreciations. The leave-one-country-out estimates (Appendix Figure A2) show that the pooled coefficient is stable to the removal of any single country except Turkey, the economy with the largest depreciations, without which it falls close to zero (Appendix Table A11). Since that economy satisfies the inclusion rule and remains in the sample, the relevant question is whether the association is present in the typical economy.

We address this question in two ways. First, we estimate the baseline model on the expanded panel of fourteen inflation-targeting emerging economies defined in Section 3.6, using the N25 network, which includes their currencies. Table 7 reports the pooled estimates, the mean-group estimates of Pesaran and Smith (1995), which allow every coefficient to differ by country and give each economy the same weight, and the permutation test. Second, we scale depreciation by the standard deviation of each country's monthly exchange rate change, so that a unit change corresponds to a one-standard-deviation depreciation in every economy and countries with volatile currencies no longer dominate by scale.

The expanded panel confirms the direction of the result and strengthens its inference. The pooled interaction is negative and highly significant at all horizons, with magnitudes about 25% smaller than in the six-country panel. With fourteen clusters, the group-level test that was underpowered in the six-country panel now rejects at $h3$ at the 5% level and at $h1$ and $h6$ at the 10% level, and the permutation test rejects at $h6$. Between 64% and 86% of the country-specific coefficients are negative. The six original economies yield almost the same estimate with the N25 network as with the main network (N17), so the change in network size does not drive the result.

The scaled-depreciation estimates point in the same direction, with an interaction of -0.746 at $h6$ in the six-country panel (wild bootstrap $p = 0.017$) and -0.541 in the fourteen-country panel ($p = 0.037$). In the thirteen-country panel of Appendix Table A11, the pooled coefficient is about one-third as large and significant at $h3$, while the mean-group estimates are virtually unchanged. We therefore read the magnitude of Table 3 as shaped by the economies with the largest depreciations, and the evidence on generality as resting on the fourteen-country panel: a smaller negative association is present in the typical inflation-targeting emerging economy.

The N25 network also allows a decomposition of received connectedness by origin. The lower block of Table 7 splits BadNet into the part received from the other emerging-market currencies and the part received from the global nodes (the broad dollar, the advanced-economy currencies, the yuan, the won, the VIX, and the uncertainty indices), each standardized, and includes both interactions with depreciation. In the fourteen-country panel, the dampening is carried by exposure to the other emerging currencies,

Table 7: Expanded panel of fourteen inflation-targeting emerging economies

	$h1$	$h3$	$h6$	$h12$
Pooled, N25 network				
14 emerging economies	-0.0570^{***}	-0.1164^{***}	-0.1266^{***}	-0.1481^{***}
	(0.0092)	(0.0144)	(0.0178)	(0.0224)
6 original economies, N25 network	-0.0678^{***}	-0.1178^{***}	-0.1230^{***}	-0.1236^{**}
	(0.0054)	(0.0076)	(0.0093)	(0.0280)
Mean group (Pesaran–Smith)				
14 economies	-0.0350^{*}	-0.0881^{**}	-0.0903^{*}	-0.0775
	(0.0181)	(0.0308)	(0.0439)	(0.0540)
Share of negative country coefficients, 14	0.79	0.86	0.71	0.64
Depreciation scaled by country volatility, pooled				
6 original economies, main network	-0.3489^{**}	-0.6756^{**}	-0.7462^{**}	-0.7688
	(0.0489)	(0.1474)	(0.1999)	(0.2059)
14 emerging economies, N25 network	-0.2375^{*}	-0.4813^{**}	-0.5414^{**}	-0.6831^{**}
	(0.0836)	(0.1457)	(0.1766)	(0.2132)
BadNet by origin, pooled, N25 network				
14 economies: exposure to other EM currencies	-0.0455^{***}	-0.0975^{***}	-0.1069^{***}	-0.1438^{***}
	(0.0121)	(0.0262)	(0.0326)	(0.0436)
14 economies: exposure to global nodes	-0.0282^{*}	-0.0517^{*}	-0.0527	-0.0300
	(0.0133)	(0.0249)	(0.0378)	(0.0603)
Permutation p-value, 14 economies (2,000 draws)			0.042	0.054
Observations, pooled 14	2,540	2,512	2,470	2,386

Note: Baseline specification estimated on Brazil, Chile, India, Mexico, South Africa, Turkey, Colombia, Peru, Indonesia, the Philippines, Thailand, Poland, Hungary, and the Czech Republic, 2010M01–2025M07. Pooled rows report CR1 standard errors in parentheses, with stars from wild cluster bootstrap p-values. Mean-group rows report the mean of the country coefficients and its standard error, with stars from a $t(G-1)$ test, equivalent to the Ibragimov and Müller (2010) test. In the scaled rows, depreciation is divided by the standard deviation of each country's monthly exchange rate change, so coefficients refer to a one-standard-deviation depreciation. In the origin rows, received connectedness is split into the part received from the other emerging-market currency nodes and the part received from all other nodes; both components are standardized and their interactions with depreciation enter jointly. Significance: $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

with an interaction between -0.045 at $h1$ and -0.144 at $h12$, all significant at the 1% level, while the interaction with global exposure is smaller and significant only at the 10% level at $h1$ and $h3$. The same pattern appears in the six original economies and, more weakly, in the thirteen-country panel, where the interaction with exposure to other emerging currencies is significant at $h3$ (Appendix Table A11). The depreciations that pass through less are therefore those shared with other emerging-market currencies. This fits both the uncertainty about persistence and the effective exchange rate channels of Section 2.4; Section 4.10 reports a direct test that separates them.

4.9. Identification Checks and Validation of BadNet

Because the design does not identify exogenous variation in exchange rates or in BadNet, this subsection reports, in Appendix Table A12, four exercises that probe the interpretation of the interaction as a conditional association that is not driven by reverse causality, pre-existing trends, or omitted variables, and documents what BadNet measures.

First, the instrumental-variable local projections, which use U.S. monetary policy surprises as a source of exchange rate variation that is external to the emerging economies,

produce negative point estimates at all horizons, but the instruments are weak (first-stage F statistics below 3) and the estimates are uninformative. With monthly data, six countries, and common shocks, U.S. monetary policy surprises do not move emerging-market exchange rates strongly enough to identify the interaction. We report the exercise for completeness and do not rely on it. Second, the Oster (2019) δ exceeds one at all horizons, and exceeds 2.5 between $h1$ and $h6$: selection on unobservables would have to be more than twice as strong as selection on the observed controls and fixed effects to eliminate the association. Third, the pre-trend placebos are null at $k = 3$ and $k = 12$ but significant at $k = 6$. We cannot exclude that depreciations in high-BadNet states tend to follow periods of somewhat lower inflation, which would bias the interaction toward negative values at medium horizons. Fourth, as shown in Table 5, the result is close to the baseline with BadNet lagged one month.

As to what BadNet measures, Appendix Table A13 shows that it is not a disguised measure of global risk. Country fixed effects explain 25% of its variance; adding the VIX and the broad dollar raises this share only to 28%, and month fixed effects to 55%, so that almost half of its variation is country-specific and time-varying. Among its determinants, the VIX enters positively, while the currency's own volatility enters negatively: received connectedness is high when the domestic currency's bad volatility is driven by the rest of the network rather than by domestic shocks. BadNet does not predict depreciation, volatility, or tail events over the following three months, so the interaction does not capture expected depreciation dynamics. BadNet also does not mechanically rise in every stress episode. Across the emerging panel, it was on average below its mean during the 2013 taper tantrum and the 2015–2016 China and commodity episode, and above its mean during the pandemic and the 2022 global tightening, which were the episodes in which emerging currencies moved most in common.

4.10. Mechanisms

This subsection confronts the results with the predictions of the channels of Section 2.4 (Table 1), using the implications that can be examined with available data; the monetary policy channel was examined in Section 4.5. Table 8 summarizes the main results, and Appendix Tables A14 and A15 report all of them.

Seven results stand out. First, BadNet flattens pass-through in both directions. Depreciations pass through less, and appreciations reduce prices less, when BadNet is high. This symmetry is what the uncertainty-about-persistence channel predicts (Table 1), and it is inconsistent with a channel that operates only through demand compression after depreciation.

Second, the lower pass-through is not limited to energy and food. It appears in core inflation, in Brazilian tradables, and, at longer horizons, in non-tradables and services. The largest coefficients are in Brazilian administered prices, which include fuel and energy tariffs whose adjustment to exchange rates is discretionary and lagged. Producer prices and industrial production show no significant association, although the sample for these variables is smaller.

Third, the financial evidence offers limited support for a credit or risk-premium channel. Sovereign spreads and private credit do not respond differently to depreciation under high BadNet, and, as shown in Section 4.5, policy rates rise less. Local equity prices fall more after high-BadNet depreciations, which is consistent with a deterioration of expected demand or profitability.

Fourth, the dampening depends on the domestic state. It is stronger when inflation is above the country's median, mirroring the high-policy-rate result, and weaker when

Table 8: Mechanisms: main results

Specification	$h1$	$h3$	$h6$	$h12$
Asymmetry				
$\Delta e^+ \times BadNet$	-0.065^{***}	-0.126^{***}	-0.123^{***}	-0.125^{***}
	(0.004)	(0.016)	(0.015)	(0.019)
$\Delta e^- \times BadNet$	0.062^{*}	0.138^{**}	0.230^{*}	0.311^{**}
	(0.024)	(0.060)	(0.113)	(0.128)
Price components, $\Delta e^+ \times BadNet$				
Core CPI (Chile, Mexico, South Africa, Turkey)	-0.082^{**}	-0.168^{*}	-0.181^{*}	-0.217^{*}
	(0.008)	(0.022)	(0.026)	(0.043)
Brazil, IPCA tradables	-0.030	-0.121^{***}	-0.154^{**}	-0.260^{***}
	(0.019)	(0.039)	(0.065)	(0.070)
Brazil, IPCA administered prices	-0.147^{***}	-0.289^{***}	-0.314^{***}	-0.349^{***}
	(0.034)	(0.058)	(0.085)	(0.071)
Monetary policy				
Change in policy rate as dependent, $\Delta e^+ \times BadNet$	-0.017^{**}	-0.044^{***}	-0.061^{**}	-0.012
	(0.005)	(0.006)	(0.012)	(0.024)
$\Delta e^+ \times BadNet\times$ policy rate above country median	-0.069^{***}	-0.089^{***}	-0.189^{***}	-0.326^{***}
	(0.004)	(0.030)	(0.033)	(0.059)
State dependence and structure, $\Delta e^+ \times BadNet \times X$				
Inflation above country median	-0.072^{**}	-0.196^{**}	-0.285^{**}	-0.408^{**}
	(0.012)	(0.020)	(0.052)	(0.077)
Credit growth above country median	0.061^{**}	0.238^{***}	0.373^{***}	0.510^{**}
	(0.014)	(0.011)	(0.034)	(0.070)
USD share of imports ($G = 7$)	0.063	0.148	0.203^{*}	0.272^{*}
	(0.023)	(0.056)	(0.062)	(0.093)
Trade network centrality, EM6	-0.201^{**}	-0.513^{**}	-0.593^{*}	-0.913
	(0.066)	(0.226)	(0.320)	(0.594)
Effective exchange rate (BIS broad NEER)				
EM6, effective depreciation × BadNet, replacing Δe^+	-0.161^{***}	-0.264^{***}	-0.293^{***}	-0.383^{***}
	(0.008)	(0.025)	(0.026)	(0.044)
EM6, $\Delta e^+ \times BadNet$, controlling for effective terms	-0.001	-0.064	-0.074	0.067
	(0.009)	(0.029)	(0.044)	(0.048)

Note: Baseline specification with the dependent variable or the additional interaction indicated in each row. Standard errors clustered by country in parentheses; stars refer to wild cluster bootstrap p-values. Brazilian components are single-country regressions with HAC standard errors and p-values. Core CPI excludes food and energy (OECD). $\Delta e^- = \max(-\Delta e, 0)$. State indicators are measured in $t-1$. Invoicing shares are from Boz et al. (2022); trade network centrality is the Katz centrality of Richmond (2019). Effective depreciation is $\max(-\Delta \ln NEER, 0) \times 100$, from the monthly-average BIS broad index; "controlling for effective terms" adds effective depreciation and its interaction with BadNet, with three lags. The complete set of results, including producer prices, activity, sovereign spreads, equity prices, and credit, is in Appendix Tables A14 and A15. Significance: $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

credit is growing above its median. The first finding suggests that the lower marginal pass-through is concentrated in periods in which firms have already raised prices and monetary policy is already restrictive. The second suggests that ample credit allows firms to pass through costs even under high network exposure.

Fifth, the dampening is smaller in economies with a larger share of dollar-invoiced imports, with p-values between 0.05 and 0.10 at $h6$ and $h12$. This is the pattern that the effective exchange rate channel predicts under dominant currency pricing (Gopinath et al., 2020; Boz et al., 2022): when import prices are set in dollars, a depreciation against the dollar raises domestic currency costs regardless of what other currencies do, leaving less room for the network state to affect pass-through.

Sixth, following Richmond (2019), we measure each economy's centrality in the global trade network by its Katz centrality, computed from bilateral trade flows scaled

by the GDP of the trading partners. Across the ten economies of the comparative panel, centrality does not change the interaction. Within the emerging panel, the dampening is stronger in years in which the economy is more central in the trade network, significantly so at $h1$ and $h3$. In the framework of Richmond (2019), central economies are more exposed to common global shocks, so this pattern is consistent with the reading that the network state matters most when the depreciation is part of a common movement.

Seventh, we test the effective exchange rate channel directly with the broad nominal effective exchange rates of the Bank for International Settlements. If high-BadNet depreciations passed through less because competitors' currencies fall at the same time, the gap between depreciation against the dollar and effective depreciation should widen with BadNet, and the interaction should weaken once depreciation is measured in effective terms. The first implication fails. The correlation between BadNet and that gap is close to zero (-0.004 in EM6 and 0.012 in EM14). The second gives a split answer. In EM6, the interaction is larger when effective depreciation replaces depreciation against the dollar (-0.29 at $h6$), and when both enter jointly only the effective interaction remains significant, so in this panel the dampening is a property of the depreciation itself, however it is measured. In the thirteen-country panel the pattern reverses (Appendix Table A11): the effective interaction is close to zero once the dollar terms are included, while the dollar interaction remains significant at $h3$ (-0.057), which is what the effective exchange rate channel predicts for the typical economy. Because the BIS indices are monthly averages and our bilateral rates are end-of-month, the two depreciation measures have a correlation of only about 0.6, and this test should be read with that caveat.

These results do not identify a single mechanism. Section 4.11 brings them together and sets each channel against the evidence.

4.11. Economic Interpretation of the Empirical Tensions

The results fit together once BadNet is read as a measure of the composition of a depreciation. When BadNet is high, the currency is moving with the network rather than on its own shocks: received connectedness rises when the currency's own volatility falls (Appendix Table A13), and the dampening is carried by exposure to the other emerging-market currencies (Section 4.8). Such depreciations pass through less to prices, and so do appreciations of the same kind.

The channels of Section 2.4 make different predictions about this pattern, and Table 9 sets each of them against the evidence. A defensive monetary response is rejected, because policy rates rise less after high-BadNet depreciations. Demand and financial contraction receives little support, because the dampening is symmetric and spreads, credit, and industrial production do not respond, although equity prices fall. The effective exchange rate channel explains the role of dollar invoicing and, in the thirteen-country panel, the result of the effective exchange rate test, but BadNet does not change the gap between bilateral and effective depreciation, so it cannot be the whole explanation. Uncertainty about persistence is consistent with all of the patterns: the symmetry, the stronger dampening when inflation and policy rates are already high and firms have recently adjusted prices, and the concentration of the effect in common emerging-market movements, whose persistence is hardest to assess at the time they occur. This is the behavior predicted by models in which firms respond quickly to idiosyncratic shocks and slowly to aggregate ones (Maćkowiak and Wiederholt, 2009), and it extends the persistence argument of Taylor (2000) from the inflation environment to the composition of the depreciation. None of these tests isolates a channel; the direct test, firms' price-setting decisions, requires microdata that are not available for these economies.

Table 9: Channels: predictions and evidence

Channel	Testable prediction	Evidence	Assessment
Uncertainty about persistence	Lower pass-through of depreciations and appreciations; stronger dampening when inflation is high	Both flattened; stronger above median inflation (Table 8); BadNet does not predict future depreciation (Table A13), so the channel must work through uncertainty rather than realized reversals	Consistent
Effective exchange rate and competitors' prices	Weaker with dollar invoicing; stronger with trade centrality; driven by exposure to other EM currencies; smaller in effective terms	Weaker with dollar invoicing; stronger with centrality; driven by EM nodes (Section 4.8); BadNet uncorrelated with the gap between dollar and effective depreciation; dampening carries over to effective terms in EM6 but stays with the dollar rate in EM13 (Table 8)	Mixed
Demand and financial contraction	Depreciation-only dampening; wider spreads, less credit, lower activity	Symmetric; spreads, credit, and industrial production do not respond; equity prices fall (Table A14)	Weak support
Defensive monetary policy	Larger policy-rate increases	Policy rates rise less (Table A7)	Rejected
Amplification	Positive interaction	Negative interaction	Rejected

Note: The channels are not mutually exclusive. The predictions are those of Table 1. "Consistent" means that the reported patterns match the predictions; none of the tests isolates a channel.

The comparison with advanced economies supports the same reading. Their pass-through is close to zero at every level of BadNet, so there is little for the network state to modify. The relevant distinction for ERPT is therefore not only between emerging and advanced economies, but between depreciations that are the currency's own and depreciations that are the network's. In the main network, what matters is exposure to the depreciation side of the network, in line with the evidence that negative and positive shocks carry different information (Black, 1976; Christie, 1982; French et al., 1987; Patton and Sheppard, 2015; Baruník et al., 2017), although Table 6 shows that this specificity is not present in every network composition.

5. Conclusion

This article asked whether a depreciation that occurs because the rest of the currency network is moving reaches consumer prices in the same way as an idiosyncratic depreciation of the same size. To answer it, we built BadNet, the share of a currency's bad volatility that is explained by shocks to other currencies and global risk factors, and used it to condition the pass-through of depreciation in panel local projections.

The answer is no. In six emerging economies, depreciations pass through less when BadNet is high. A one-standard-deviation increase in BadNet lowers the one-month pass-through of a 1% depreciation by about as much as the pass-through at the mean, and the six-month pass-through by about two-thirds. The association is close to zero in advanced economies, where pass-through itself is small, and it is present, with a smaller magnitude, in a panel of fourteen inflation-targeting emerging markets.

The pattern of the evidence points to an explanation. The dampening is symmetric

across depreciations and appreciations, is carried by exposure to other emerging-market currencies, is stronger when inflation is already high, and does not come from a stronger monetary response or tighter financial conditions. It is most consistent with firms' uncertainty about the persistence of movements they share with the rest of the network, which gives them reason to wait before adjusting prices. The alternative that common depreciations simply change relative prices less receives mixed support.

Five limits must be stated with equal clarity. First, the magnitude of the effect in the six-country panel is shaped by the economies with the largest depreciations, which is why the evidence on generality rests on the fourteen-country panel. Second, over the full sample, the persistence of the association beyond the first months and its specificity to the bad-volatility layer depend on the monthly uncertainty indices for China and Russia; where daily proxies exist, the result does not depend on how these nodes are treated, but outside that period it cannot be verified. Third, with six clusters no cluster-robust test is exactly sized, and the most conservative procedures reject only in the expanded panels of fourteen emerging and seven advanced economies. Fourth, U.S. monetary policy surprises are too weak an instrument to identify the interaction, and one of the three pre-trend placebos is significant, so the estimates remain conditional associations rather than causal effects. Fifth, the network measure is sensitive to the numeraire, which suggests, without proving, that the dollar-based network is the relevant one for emerging-market prices.

For policy, the results suggest that central banks in emerging economies can monitor not only how much their currency depreciates, but how much of the movement is shared with the network, because a depreciation that occurs with the network is likely to pass through less than one of the same size that occurs alone. Given the limits above, BadNet is best used as a monitoring indicator rather than as a basis for calibrating policy responses. Future work could test the uncertainty-about-persistence channel directly with firm-level prices and survey expectations, ask whether BadNet improves out-of-sample inflation forecasts, and extend the analysis to economies with quarterly consumer prices through mixed-frequency methods.

Appendix A. Data and Additional Results

Appendix A.1. Data

Table A1: GJR-GARCH(1,1) parameters and share of depreciation days by node

Node	α	β	γ	p-value of γ	Persistence	Share of depreciation days
BRL	0.125	0.900	−0.071	< 0.001	0.990	0.508
USD	0.055	0.957	−0.033	< 0.001	0.996	0.497
JPY	0.038	0.948	0.016	0.078	0.995	0.483
TRY	0.138	0.873	−0.024	0.080	0.999	0.435
VIX	0.247	0.751	−0.307	< 0.001	0.845	0.490
MXN	0.141	0.894	−0.099	< 0.001	0.985	0.513
ZAR	0.064	0.950	−0.055	< 0.001	0.987	0.463
CLP	0.083	0.933	−0.037	< 0.001	0.998	0.498
CNY	0.086	0.900	0.026	0.012	0.999	0.514
KRW	0.071	0.951	−0.049	< 0.001	0.998	0.499
INR	0.061	0.955	−0.033	< 0.001	0.999	0.483
EUR	0.042	0.965	−0.017	0.009	0.999	0.496
CAD	0.051	0.960	−0.026	< 0.001	0.997	0.479
AUD	0.050	0.956	−0.028	< 0.001	0.993	0.506
EPU_US	0.015	0.970	0.017	0.020	0.994	0.494
EPU_CH	0.022	0.001	0.970	< 0.001	0.509	0.906
EPU_RU	0.015	0.001	0.973	< 0.001	0.502	0.829
Additional nodes of the N21 network						
GBP	0.044	0.963	−0.027	< 0.001	0.994	0.507
CHF	0.028	0.963	0.006	0.448	0.994	0.493
SEK	0.040	0.964	−0.015	0.030	0.996	0.489
NOK	0.049	0.957	−0.024	0.002	0.995	0.490

Note: ARMA(1,1) mean equation, Student-t innovations. The asymmetry term γ multiplies squared negative innovations; because positive innovations correspond to depreciation, a stronger dollar, or rising risk and uncertainty, $\gamma < 0$ means that conditional variance responds more to shocks in the depreciation direction. The share of days of depreciation pressure is the share of days with $z_t > 0$. Persistence is $\alpha + \beta + \gamma\kappa$, where κ is the probability of a negative standardized innovation under the estimated Student-t distribution.

Table A2: Data dictionary of the network nodes

Node	Series	Source	Transformation
Main network, N17 (17 nodes), 2006-01-03 to 2026-06-01			
BRL	Brazilian real per U.S. dollar	FRED, DEXBZUS	log return
MXN	Mexican peso per U.S. dollar	FRED, DEXMXUS	log return
CLP	Observed dollar rate, pesos per U.S. dollar	Banco Central de Chile	lagged one business day; log return
ZAR	South African rand per U.S. dollar	FRED, DEXSFUS	log return
TRY	Turkish lira per U.S. dollar	Yahoo Finance, TRY=X	log return
INR	Indian rupee per U.S. dollar	FRED, DEXINUS	log return
CNY	Chinese yuan per U.S. dollar	FRED, DEXCHUS	log return
KRW	Korean won per U.S. dollar	FRED, DEXKOUS	log return
EUR	Euro, inverted from U.S. dollars per euro	FRED, DEXUSEU	inverted; log return
JPY	Japanese yen per U.S. dollar	FRED, DEXJPUS	log return
CAD	Canadian dollar per U.S. dollar	FRED, DEXCAUS	log return
AUD	Australian dollar, inverted from U.S. dollars per AUD	FRED, DEXUSAL	inverted; log return
USD	Nominal broad U.S. dollar index	FRED, DTWEXBGS	log return
VIX	CBOE volatility index	FRED, VIXCLS	log change
EPU_US	Daily economic policy uncertainty, United States	FRED, USEPUINDXD; Baker et al. (2016)	log change
EPU_CH	Monthly economic policy uncertainty, China	policyuncertainty.com	repeated on business days; log change
EPU_RU	Monthly economic policy uncertainty, Russia	policyuncertainty.com	repeated on business days; log change
Daily proxies for China and Russia, 2013-11-18 to 2022-02-11			
VXFXI	CBOE China ETF volatility index	FRED, VXFXICLS	log change
RVI	Russian volatility index	Moscow Exchange	log change
Additional advanced-economy nodes of N21, 2006-01-03 to 2026-06-01			
GBP	Pound sterling, inverted from U.S. dollars per pound	FRED, DEXUSUK	inverted; log return
CHF	Swiss franc per U.S. dollar	FRED, DEXSZUS	log return
SEK	Swedish krona per U.S. dollar	FRED, DEXSDUS	log return
NOK	Norwegian krone per U.S. dollar	FRED, DEXNOUS	log return
Additional emerging-economy nodes of N25, to 2025-07			
8 nodes	COP, PEN, IDR, PHP, THB, PLN, HUF, CZK, domestic currency per U.S. dollar	BIS daily exchange rates	log return

Note: All series are aligned on a business-day calendar; missing values due to national holidays are filled with the previous observation. Every node is oriented so that an increase indicates depreciation pressure (depreciation of the domestic currency, a stronger dollar, higher risk aversion, or higher uncertainty).

Table A3: Descriptive statistics of the emerging-economy panel

		Inflation		Exchange rate			BadNet (FROM^B)			Policy
Country	Months	Mean	SD	$\overline{\Delta e^+}$	Share > 0	SD Δe	Mean	P10	P90	rate
Brazil	195	5.62	0.36	1.96	0.51	4.32	54.2	34.2	70.5	10.01
Chile	195	4.04	0.37	1.55	0.52	3.60	57.1	41.5	72.2	4.15
India	195	5.50	0.69	0.91	0.58	2.14	53.1	29.4	74.6	6.64
Mexico	195	4.31	0.36	1.33	0.45	3.49	59.5	42.8	73.9	6.15
South Africa	195	4.84	0.37	1.87	0.48	4.19	61.0	46.2	76.2	6.07
Turkey	195	19.18	1.92	2.49	0.68	4.76	35.8	15.3	63.3	14.33
All	1,170	7.25	0.99	1.69	0.54	3.87	53.4	26.9	73.1	7.89

Note: 2010M01–2026M03. Inflation mean is the average monthly log change of the CPI multiplied by 12 (annualized, %); inflation SD is the standard deviation of the monthly change (%). $\overline{\Delta e^+}$ is the average of $\max(\Delta e, 0)$ in % per month, "Share > 0" the share of months with depreciation, and SD Δe the standard deviation of the monthly exchange rate change. BadNet is the monthly average of received connectedness in the bad-volatility network, before standardization. Policy rate is the sample mean, in % per year. All series are available for every month for all six countries, except two missing policy-rate observations for Chile; the estimation samples in Table 3 fall from 1,146 to 1,074 observations only because the cumulative outcome requires future prices.

Table A4: Data blocks and sample treatment across empirical stages

Data block or group	Variables or units	Empirical use
Currency network, N17 (17 nodes)	BRL, MXN, CLP, ZAR, TRY, INR, CNY, KRW, EUR, JPY, CAD, AUD, broad U.S. dollar index; VIX; EPU United States (daily); EPU China and EPU Russia (monthly). Daily, 2006-01-03 to 2026-06-01	Construction of BadNet and measurement of bad-volatility spillovers.
Alternative networks	21 nodes (main network plus GBP, CHF, SEK, NOK); 15 nodes (without EPU China and Russia); 17 nodes with daily proxies VXFXI and RVI (2013-11 to 2022-02); 25 nodes (main network plus COP, PEN, IDR, PHP, THB, PLN, HUF, CZK, to 2025-07); 23 nodes (25 without the monthly EPUs)	Sensitivity to network composition and expanded panel.
Emerging ERPT panel	Brazil, Mexico, Chile, South Africa, Turkey, India; 2010M01–2026M03	Baseline pass-through regressions for economies with greater exchange rate vulnerability.
Advanced ERPT panel	Euro area, Japan, Canada (AE3); AE7 adds United Kingdom, Switzerland, Sweden, Norway	Comparative pass-through regressions for advanced monetary regimes.
Expanded emerging panel	Baseline six plus Colombia, Peru, Indonesia, Philippines, Thailand, Poland, Hungary, Czech Republic	Generalization; inclusion rule: inflation targeting and floating exchange rate.
Special cases	United States; South Korea; Australia; China	In the network. U.S. and Korea used only in sensitivity exercises; Australia excluded from the panels (quarterly CPI); China excluded from the panels (no inflation targeting, managed exchange rate).
Baseline outcome	Cumulative CPI change, $100[\ln CPI_{t+h} - \ln CPI_{t-1}]$	Main dependent variable in the pass-through regressions.
Main ERPT variable	Positive exchange rate depreciation, $\max(\Delta e_{i,t}, 0)$	Main exchange rate pass-through variable in the local projections.
Network conditioning variable	Monthly average of $\text{FROM}^{B}_{i,t}$, standardized on the emerging panel	State variable conditioning the marginal effect of depreciation on inflation.
Baseline controls	Three lags of monthly inflation, depreciation, BadNet, and their interaction; country and month fixed effects	Dynamics and all common global shocks (absorbed by month fixed effects).
Additional controls and outcomes	Policy rate; VIX, broad dollar and risk-off interactions; core CPI; Brazilian IPCA components; PPI; industrial production; short-term rate; sovereign spread; equity prices; private credit; dollar invoicing shares	Robustness and mechanism analysis.
Instruments	U.S. monetary policy surprises (Jarociński and Karadi, 2020)	Instrumental-variable local projections.

Appendix A.2. Additional results

Table A5: Conservative inference in the emerging-economy panel

Procedure	$h0$	$h1$	$h3$	$h6$	$h12$
$\Delta e^{+} \times BadNet$	−0.0065	−0.0752	−0.1522	−0.1684	−0.1873
p-values					
CR1, $t(G-1)$	0.204	< 0.001	0.001	0.001	< 0.001
Wild cluster bootstrap (Webb)	0.133	0.003	0.012	0.011	0.011
Driscoll–Kraay	0.328	< 0.001	0.005	0.035	0.069
Ibragimov–Müller	0.540	0.150	0.132	0.249	0.219
Wild bootstrap, Holm (13 horizons)	0.133	0.040	0.132	0.132	0.132
Wild bootstrap, Benjamini–Hochberg	0.133	0.015	0.015	0.015	0.015
Permutation across countries (720)				0.092	0.128
95% sup-t simultaneous band					
Lower bound	−0.023	−0.089	−0.225	−0.255	−0.259
Upper bound	0.010	−0.062	−0.079	−0.082	−0.116
Two-step bootstrap (rolling-window BadNet)					
Coefficient				−0.208	−0.243
Inflation of the standard deviation				1.91	2.40
95% interval, lower bound				−0.260	−0.316
95% interval, upper bound				−0.077	−0.081
Monte Carlo rejection rate under the null at 5% (500 replications)					
CR1, $t(G-1)$				0.194	
CR1, normal				0.322	
Wild cluster bootstrap				0.126	
Ibragimov–Müller				0.058	

Note: Baseline specification, EM6 panel, $G = 6$. The Ibragimov–Müller test applies a $t(5)$ test to the six country-level coefficients. Holm and Benjamini–Hochberg adjustments are computed over the wild bootstrap p-values of the thirteen horizons. The permutation test reassigns the BadNet series across the six countries (all $6! = 720$ permutations). The two-step bootstrap re-estimates a rolling-window version of BadNet in each of 199 replications; its correlation with the TVP-VAR measure is 0.52. In the Monte Carlo experiment, run at $h6$, the data are generated from the baseline model with the interaction set to zero and residuals resampled in 12-month blocks within country.

Table A6: Frequency decomposition of BadNet, emerging economies

	$h0$	$h1$	$h3$	$h6$	$h12$
Baseline network, $H = 10$, cut at $\pi/2$					
Short-run	-0.0125	-0.0655	-0.1203^{*}	-0.1485	-0.1406
	(0.0048)	(0.0186)	(0.0302)	(0.0549)	(0.0403)
Persistent	-0.0035	-0.0765^{***}	-0.1556^{**}	-0.1680^{**}	-0.2039^{**}
	(0.0046)	(0.0077)	(0.0290)	(0.0314)	(0.0420)
$H = 100$					
Total BadNet	-0.0066	-0.0748^{***}	-0.1441^{***}	-0.1555^{***}	-0.1608^{***}
	(0.0044)	(0.0036)	(0.0166)	(0.0193)	(0.0158)
Short-run, cut at $\pi/2$	-0.0146^{*}	-0.0631^{*}	-0.0900^{*}	-0.0893	-0.0194
	(0.0045)	(0.0156)	(0.0239)	(0.0383)	(0.0892)
Persistent, cut at $\pi/2$	-0.0011	-0.0752^{**}	-0.1532^{**}	-0.1649^{**}	-0.2047^{*}
	(0.0045)	(0.0082)	(0.0295)	(0.0309)	(0.0520)
Short-run, cut at $\pi/5$	-0.0136^{*}	-0.0726^{**}	-0.1231^{**}	-0.1389^{*}	-0.1057
	(0.0046)	(0.0130)	(0.0209)	(0.0350)	(0.0505)
Persistent, cut at $\pi/5$	0.0023	-0.0651^{**}	-0.1330^{*}	-0.1344^{*}	-0.1885
	(0.0052)	(0.0133)	(0.0386)	(0.0405)	(0.0835)
Short-run, cut at $\pi/22$	-0.0134^{**}	-0.0771^{***}	-0.1366^{**}	-0.1510^{**}	-0.1434^{**}
	(0.0047)	(0.0074)	(0.0172)	(0.0286)	(0.0279)
Persistent, cut at $\pi/22$	0.0030	-0.0639^{*}	-0.1349	-0.1392	-0.1848
	(0.0058)	(0.0200)	(0.0563)	(0.0532)	(0.0993)

Note: Each row reports the coefficient on $\Delta e^{+} \times BadNet^{k}$ from a separate baseline regression in which the standardized component k replaces aggregate BadNet. Standard errors clustered by country in parentheses; stars refer to wild cluster bootstrap p-values. Under the labeling convention of Section 3.2, the cuts at $\pi/2$, $\pi/5$, and $\pi/22$ separate cycles of up to 2, 5, and 22 trading days. Significance: $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

Table A7: Regime mechanism: BadNet during high-interest-rate periods

	$h0$	$h1$	$h3$	$h6$	$h12$
$\Delta e^{+} \times BadNet \times HighPolicy$					
At or above country median (baseline)	-0.0094^{*}	-0.0687^{***}	-0.0894^{***}	-0.1889^{***}	-0.3256^{***}
	(0.0046)	(0.0045)	(0.0304)	(0.0325)	(0.0588)
At or above country 75th percentile	-0.0239	0.0335	0.0310	-0.1175	-0.2332
	(0.0143)	(0.0227)	(0.0447)	(0.0968)	(0.1346)
Standardized policy rate (continuous)	-0.0187	-0.0292	-0.0326	-0.0814^{**}	-0.1967^{**}
	(0.0099)	(0.0168)	(0.0239)	(0.0325)	(0.0751)
Change in the policy rate as dependent variable					
$\Delta e^{+} \times BadNet$	-0.0080	-0.0173^{**}	-0.0441^{***}	-0.0614^{**}	-0.0117
	(0.0057)	(0.0045)	(0.0065)	(0.0123)	(0.0242)

Note: EM6 panel. *HighPolicy* is measured in $t-1$ relative to each country's own distribution of the policy rate. Standard errors clustered by country in parentheses; stars refer to wild cluster bootstrap p-values. In the last row the dependent variable is the cumulative change in the policy rate between $t-1$ and $t+h$. Significance: $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

Table A8: Robustness of the interaction between depreciation and BadNet: complete set of specifications

Specification	$h1$	$h3$	$h6$	$h12$
Baseline	-0.075^{***}	-0.152^{**}	-0.168^{**}	-0.187^{**}
	(0.004)	(0.019)	(0.023)	(0.019)
Timing and aggregation of BadNet				
BadNet lagged $(t-1)$	-0.072^{***}	-0.139^{**}	-0.130^{***}	-0.161^{***}
	(0.004)	(0.017)	(0.010)	(0.017)
Monthly maximum	-0.065^{***}	-0.125^{***}	-0.110^{***}	-0.094^{**}
	(0.002)	(0.009)	(0.012)	(0.035)
End of month	-0.077^{***}	-0.140^{***}	-0.148^{***}	-0.142^{**}
	(0.009)	(0.019)	(0.023)	(0.021)
Outcome and dynamics				
Year-on-year inflation at $t+h$	-0.091^{**}	-0.199^{**}	-0.213^{**}	-0.178^{**}
	(0.023)	(0.050)	(0.039)	(0.020)
Twelve lags	-0.084^{***}	-0.171^{***}	-0.195^{**}	-0.191^{**}
	(0.010)	(0.026)	(0.040)	(0.033)
Monthly inflation at $t+h$ (non-cumulative)	-0.068^{***}	-0.015^{**}	0.009^{**}	0.006
	(0.007)	(0.004)	(0.002)	(0.006)
Balanced constant sample	-0.078^{***}	-0.156^{***}	-0.173^{***}	-0.187^{***}
	(0.004)	(0.020)	(0.025)	(0.019)
Country FE and global controls, no month FE	-0.079^{***}	-0.170^{**}	-0.207^{*}	-0.274^{**}
	(0.009)	(0.030)	(0.041)	(0.045)
Additional controls				
+ squared depreciation, $(\Delta e^{+})^2$	-0.052^{**}	-0.138^{***}	-0.145^{***}	-0.154^{***}
	(0.011)	(0.025)	(0.023)	(0.017)
+ $(\Delta e^{+})^2$ and $(\Delta e^{+})^2 \times BadNet$: linear interaction	0.041	-0.033	-0.023	0.040
	(0.021)	(0.024)	(0.035)	(0.065)
quadratic interaction, $(\Delta e^{+})^2 \times BadNet$	-0.0079^{**}	-0.0089^{**}	-0.0102^{**}	-0.0163^{**}
	(0.0018)	(0.0034)	(0.0038)	(0.0070)
+ $\Delta e^{+}\times$ own FX volatility $(t-1)$	-0.073^{***}	-0.152^{***}	-0.159^{**}	-0.178^{***}
	(0.004)	(0.017)	(0.019)	(0.015)
+ lagged policy rate	-0.079^{***}	-0.158^{***}	-0.176^{**}	-0.193^{***}
	(0.005)	(0.021)	(0.026)	(0.022)
+ $\Delta e^{+}\times$ risk-off (VIX $\geq$ p80)	-0.055^{***}	-0.120^{**}	-0.128^{***}	-0.106^{*}
	(0.006)	(0.015)	(0.017)	(0.052)
+ $\Delta e^{+}\times$ VIX and $\Delta e^{+}\times$ dollar	-0.063^{***}	-0.142^{***}	-0.162^{***}	-0.183^{***}
	(0.006)	(0.017)	(0.021)	(0.018)
+ $\Delta e^{+}\times$ EPU U.S., VIX, and dollar	-0.062^{***}	-0.141^{**}	-0.162^{***}	-0.177^{**}
	(0.005)	(0.017)	(0.020)	(0.019)
Winsorized at 1/99%	-0.050^{**}	-0.104^{**}	-0.126^{*}	-0.169^{*}
	(0.008)	(0.027)	(0.040)	(0.045)
Numeraire (alternative network measures in Table A9)				
Euro numeraire	-0.039^{**}	-0.016	0.022	0.091
	(0.019)	(0.041)	(0.045)	(0.076)
Good volatility layer				
Placebo: good-volatility network	-0.047^{**}	-0.021	0.024	0.057
	(0.016)	(0.058)	(0.090)	(0.116)
Both layers: bad volatility	-0.105^{**}	-0.315^{**}	-0.410	-0.486^{**}
	(0.017)	(0.075)	(0.103)	(0.099)
Both layers: good volatility	0.052	0.271	0.408	0.511
	(0.022)	(0.099)	(0.148)	(0.155)

Note: Coefficient on the interaction between depreciation and the indicated network measure. Standard errors clustered by country (CR1) in parentheses; stars refer to wild cluster bootstrap p-values (Webb weights, null imposed; 1,999 replications for the squared-depreciation and own-volatility rows). Unless stated otherwise, regressions follow the baseline specification of Table 3. "Both layers" includes the bad-volatility and good-volatility interactions in the same regression; the euro numeraire network re-estimates all nodes against the euro; own FX volatility is the standard deviation of monthly depreciation over the previous twelve months. Significance: $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

Table A9: Alternative measures of network exposure, emerging economies

Measure	$h1$	$h3$	$h6$	$h12$	Corr.
Baseline: FROM^B, bad volatility	-0.075^{***}	-0.152^{**}	-0.168^{**}	-0.187^{**}	1.00
	(0.004)	(0.019)	(0.023)	(0.019)	
Other measures from the bad-volatility network					
TO^B, transmitted connectedness	-0.055^{***}	-0.146^{**}	-0.165^{*}	-0.208^{*}	0.38
	(0.008)	(0.039)	(0.051)	(0.071)	
$\text{NET}^B = \text{TO}^B - \text{FROM}^B$	-0.019^{**}	-0.085^{*}	-0.101	-0.149	-0.15
	(0.008)	(0.036)	(0.050)	(0.083)	
TCI^B (global state)	-0.024^{*}	-0.064	-0.041	-0.097	
	(0.013)	(0.038)	(0.047)	(0.090)	
External state (mean of other EMs)	0.013^{*}	-0.029	-0.014	-0.062	
	(0.008)	(0.060)	(0.094)	(0.123)	
FROM *in networks of total volatility*					
GJR-GARCH variance, both layers	-0.051^{**}	-0.106^{*}	-0.152^{**}	-0.184^{*}	0.53
	(0.007)	(0.021)	(0.030)	(0.040)	
Symmetric sGARCH variance	0.013	0.080	0.118	0.212	0.39
	(0.012)	(0.044)	(0.060)	(0.111)	

Note: Coefficient on the interaction between depreciation and the indicated measure, each standardized with the emerging-economy moments, in the baseline specification of Table 3, EM6 panel. Standard errors clustered by country (CR1) in parentheses; stars refer to wild cluster bootstrap p-values (Webb weights, null imposed, 1,999 replications). The total-volatility networks use the same 17 nodes and TVP-VAR specification as the main network. The last column (Corr.) is the correlation of the monthly measure with BadNet in the EM6 panel. Significance: $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

Table A10: Country-level estimates of the interaction (Ibragimov–Müller)

	$h0$	$h1$	$h3$	$h6$	$h12$
Brazil	-0.009	-0.045	-0.115	-0.166	-0.238
Chile	0.004	0.021	0.031	0.051	0.072
India	0.003	-0.040	-0.107	-0.011	0.047
Mexico	-0.003	-0.024	-0.034	-0.001	-0.069
South Africa	0.008	0.013	0.016	-0.001	-0.034
Turkey	0.008	-0.063	-0.248	-0.289	-0.323
Mean	0.002	-0.023	-0.076	-0.069	-0.091
p-value, $t(5)$	0.540	0.150	0.132	0.249	0.219

Note: Baseline specification estimated by OLS separately for each country. Because month fixed effects cannot be estimated within a single country, they are replaced by observable global controls (the VIX and the monthly changes in oil prices, the broad dollar, global food prices, the ten-year Treasury yield, and the federal funds rate).

Table A11: Estimates excluding Turkey

Specification	$h1$	$h3$	$h6$	$h12$
$\Delta e^+ \times BadNet$, *baseline specification*				
EM5, main network	−0.032	−0.044	−0.024	−0.002
	(0.023)	(0.035)	(0.052)	(0.059)
EM13, N25 network, pooled	−0.014	−0.047**	−0.045	−0.055
	(0.011)	(0.017)	(0.026)	(0.038)
EM13, mean group	−0.033	−0.088**	−0.090*	−0.061
	(0.019)	(0.033)	(0.047)	(0.056)
Depreciation scaled by country volatility				
EM5, main network	−0.124	−0.151	−0.040	0.080
	(0.083)	(0.117)	(0.166)	(0.214)
EM13, N25 network	−0.054	−0.160**	−0.160*	−0.218
	(0.039)	(0.063)	(0.087)	(0.130)
EM13, BadNet by origin				
Exposure to other EM currencies	−0.013	−0.037**	−0.038	−0.052
	(0.009)	(0.014)	(0.022)	(0.029)
EM13, effective exchange rate				
Effective depreciation × BadNet, replacing Δe^+	−0.012	−0.035	−0.042	−0.058
	(0.014)	(0.023)	(0.030)	(0.046)
Both jointly: $\Delta e^+ \times BadNet$	−0.018	−0.057***	−0.049	−0.052
	(0.012)	(0.018)	(0.027)	(0.035)
Both jointly: effective × BadNet	0.008	0.024	0.005	−0.013
	(0.015)	(0.023)	(0.029)	(0.044)

Note: EM5 is the EM6 panel without Turkey; EM13 is the expanded panel without Turkey. Pooled rows report CR1 standard errors in parentheses with stars from wild cluster bootstrap p-values; mean-group rows report the mean of the country coefficients with stars from a $t(G-1)$ test. Other definitions as in Tables 7 and 8. Significance: $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

Table A12: Identification checks, emerging economies

	$h0$	$h1$	$h3$	$h6$	$h12$
LP-IV, instruments: U.S. monetary policy surprise and its product with lagged BadNet					
$\Delta e^+ \times BadNet$	−0.079	−0.147	−0.272	−0.623	−1.404
	(0.154)	(0.423)	(0.589)	(0.758)	(1.568)
First-stage F, Δe^+	1.4	1.4	1.4	1.5	1.6
First-stage F, $\Delta e^+ \times BadNet$	2.4	2.7	2.7	2.8	3.0
Comparable OLS (no month FE)	−0.005	−0.079	−0.170	−0.207	−0.274
Sensitivity to unobservables (Oster, 2019)					
δ that drives the interaction to zero	1.12	3.92	3.94	2.71	1.51
Pre-trend placebos, $100[\ln CPI_{t-1} - \ln CPI_{t-1-k}]$*, with* $k = h$					
$\Delta e^+ \times BadNet$			−0.003	−0.058***	0.003
			(0.019)	(0.014)	(0.046)

Note: In the LP-IV, depreciation and its interaction with BadNet are instrumented by the Jarociński and Karadi (2020) monetary policy surprise and by the surprise interacted with lagged BadNet; first-stage F statistics are cluster-robust Wald statistics for the two excluded instruments divided by two; the regression includes country fixed effects, observable global controls, and the baseline lags. CR1 standard errors clustered by country in parentheses, p-values with $t(G-1)$ critical values for stars, $G = 6$. The Oster δ uses $R_{\max} = 1.3\tilde{R}^2$. For the placebos, CR1 standard errors in parentheses and stars from wild cluster bootstrap p-values. Significance: $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

Table A13: What BadNet measures: variance decomposition, determinants, and stress episodes

Share of the variance of monthly BadNet explained (R^2), emerging panel			
Country fixed effects		0.246	
+ VIX and broad dollar		0.276	
+ TCI^B		0.481	
Country and month fixed effects		0.551	
Determinants of BadNet (standardized regressors, country FE)			
	Coefficient	Std. error	p-value, CR1, $t(G-1)$
VIX	0.145	0.047	0.028
EPU United States	0.132	0.069	0.112
Broad dollar	0.008	0.031	0.798
Inflation $(t-1)$	−0.005	0.005	0.437
Policy rate $(t-1)$	0.007	0.014	0.657
Own exchange rate volatility $(t-1)$	−0.146	0.017	< 0.001
Does BadNet predict the next three months? (coefficient on BadNet)			
	Coefficient	Std. error	p-value
Cumulative depreciation	0.043	0.091	0.657
Exchange rate volatility	−0.093	0.067	0.228
Tail event indicator	−0.017	0.027	0.560
Mean standardized BadNet in stress episodes			
Taper tantrum (2013)		−0.35	
China and commodity prices (2015–2016)		−0.66	
COVID-19 (2020)		0.27	
Global tightening (2022)		0.39	
Other months		0.01	

Note: EM6 panel, 2010M01–2026M03. The determinants regression has $R^2 = 0.354$ and 1,097 observations. The predictive regressions have 1,152 observations, and adding BadNet raises their R^2 by less than 0.002. Episode windows: taper tantrum, 2013M05–2013M09; China and commodity prices, 2015M08–2016M02; COVID-19, 2020M02–2020M05; global tightening, 2022M03–2022M10.

Table A14: Mechanisms: complete set of price, financial, and state-dependence results

Specification	$h1$	$h3$	$h6$	$h12$
Asymmetry: separate depreciation and appreciation terms				
$\Delta e^+ \times BadNet$	-0.065^{***}	-0.126^{***}	-0.123^{***}	-0.125^{***}
	(0.004)	(0.016)	(0.015)	(0.019)
$\Delta e^- \times BadNet$	0.062^{*}	0.138^{**}	0.230^{*}	0.311^{**}
	(0.024)	(0.060)	(0.113)	(0.128)
Price components, $\Delta e^+ \times BadNet$				
Core CPI (Chile, Mexico, South Africa, Turkey)	-0.082^{**}	-0.168^{*}	-0.181^{*}	-0.217^{*}
	(0.008)	(0.022)	(0.026)	(0.043)
Brazil, IPCA tradables	-0.030	-0.121^{***}	-0.154^{**}	-0.260^{***}
	(0.019)	(0.039)	(0.065)	(0.070)
Brazil, IPCA non-tradables	0.001	-0.010	-0.086^{*}	-0.158^{**}
	(0.017)	(0.026)	(0.047)	(0.062)
Brazil, IPCA services	-0.004	-0.020	-0.082^{**}	-0.140^{**}
	(0.013)	(0.021)	(0.042)	(0.063)
Brazil, IPCA administered prices	-0.147^{***}	-0.289^{***}	-0.314^{***}	-0.349^{***}
	(0.034)	(0.058)	(0.085)	(0.071)
PPI (four countries)	0.125	0.110	-0.005	-0.037
	(0.065)	(0.049)	(0.046)	(0.091)
Industrial production (five countries)	0.066	-0.007	0.036	0.033
	(0.127)	(0.095)	(0.046)	(0.058)
Financial variables as dependent, $\Delta e^+ \times BadNet$				
Short-term interbank rate (p.p.)	-0.019^{**}	-0.013	0.007	0.056^{**}
	(0.006)	(0.011)	(0.017)	(0.027)
Sovereign spread (p.p.)	-0.003	-0.001	-0.003	-0.024
	(0.006)	(0.012)	(0.019)	(0.022)
Local equity index (log, %)	-0.196	-0.328^{**}	-0.484^{*}	-0.720^{*}
	(0.085)	(0.109)	(0.153)	(0.189)
Private credit (% of GDP)	-0.012	0.028	0.062	0.130
	(0.020)	(0.041)	(0.058)	(0.056)
State dependence, $\Delta e^+ \times BadNet \times State_{t-1}$				
Inflation above country median	-0.072^{**}	-0.196^{**}	-0.285^{**}	-0.408^{**}
	(0.012)	(0.020)	(0.052)	(0.077)
FX volatility above country median	-0.021	0.014	-0.016	0.077
	(0.022)	(0.056)	(0.053)	(0.056)
Credit growth above country median	0.061^{**}	0.238^{***}	0.373^{***}	0.510^{**}
	(0.014)	(0.011)	(0.034)	(0.070)

Note: Baseline specification with the dependent variable or the additional interaction indicated in each row. Standard errors clustered by country in parentheses; stars refer to wild cluster bootstrap p-values. Brazilian components are single-country regressions with HAC standard errors and HAC p-values. Core CPI excludes food and energy (OECD). $\Delta e^- = \max(-\Delta e, 0)$ is the appreciation component. Invoicing, centrality, and effective exchange rate results are in Appendix Table A15. Significance: $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

Table A15: Mechanisms: invoicing, trade centrality, and the effective exchange rate

Specification	$h1$	$h3$	$h6$	$h12$
Dollar invoicing, $\Delta e^+ \times BadNet \times USD\ share$				
USD share of imports ($G = 7$)	0.063	0.148	0.203^{*}	0.272^{*}
	(0.023)	(0.056)	(0.062)	(0.093)
USD share of trade ($G = 9$)	0.049	0.107	0.146^{*}	0.191^{*}
	(0.013)	(0.029)	(0.035)	(0.046)
Trade network centrality (Richmond, 2019), $\Delta e^+ \times BadNet \times Centrality$				
Comparative panel ($G = 10$)	0.021	0.076	0.091	0.153
	(0.036)	(0.073)	(0.090)	(0.126)
Emerging panel, centrality in $t-1$ ($G = 6$)	-0.201^{**}	-0.513^{**}	-0.593^{*}	-0.913
	(0.066)	(0.226)	(0.320)	(0.594)
Effective exchange rate: effective depreciation $\times$ *BadNet replacing* $\Delta e^+ \times BadNet$				
EM6	-0.161^{***}	-0.264^{***}	-0.293^{***}	-0.383^{***}
	(0.008)	(0.025)	(0.026)	(0.044)
EM14	-0.124^{***}	-0.183^{***}	-0.208^{***}	-0.224^{***}
	(0.026)	(0.032)	(0.036)	(0.032)
Effective exchange rate: both interactions jointly				
EM6, $\Delta e^+ \times BadNet$	-0.001	-0.064	-0.074	0.067
	(0.009)	(0.029)	(0.044)	(0.048)
EM6, effective $\times$ BadNet	-0.153^{**}	-0.177^{**}	-0.202^{**}	-0.531^{**}
	(0.021)	(0.042)	(0.060)	(0.110)
EM14, $\Delta e^+ \times BadNet$	-0.014^{*}	-0.068^{***}	-0.062^{***}	-0.081^{***}
	(0.009)	(0.014)	(0.017)	(0.014)
EM14, effective $\times$ BadNet	-0.098^{*}	-0.107	-0.146	-0.143^{**}
	(0.034)	(0.045)	(0.052)	(0.041)

Note: Baseline specification with the additional interaction indicated in each row. Standard errors clustered by country in parentheses; stars refer to wild cluster bootstrap p-values. Invoicing shares are from Boz et al. (2022), for the countries of the comparative panel with available data. Katz centrality is computed from annual bilateral trade flows (IMF Direction of Trade Statistics) scaled by GDP. Effective depreciation uses the monthly-average BIS broad nominal effective exchange rate; EM14 uses the N25 network. Significance: $^{***}p < 0.01$; $^{**}p < 0.05$; $^{*}p < 0.10$.

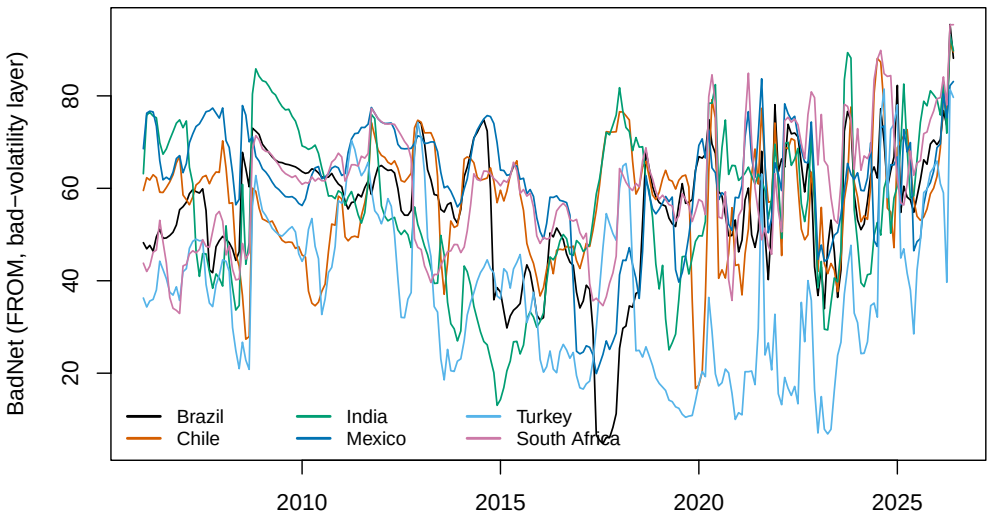


Figure A1: Monthly BadNet by emerging economy

Note: Monthly average of received connectedness in the bad-volatility network, $\text{FROM}^{B}_{i,t}$, main network (N17).

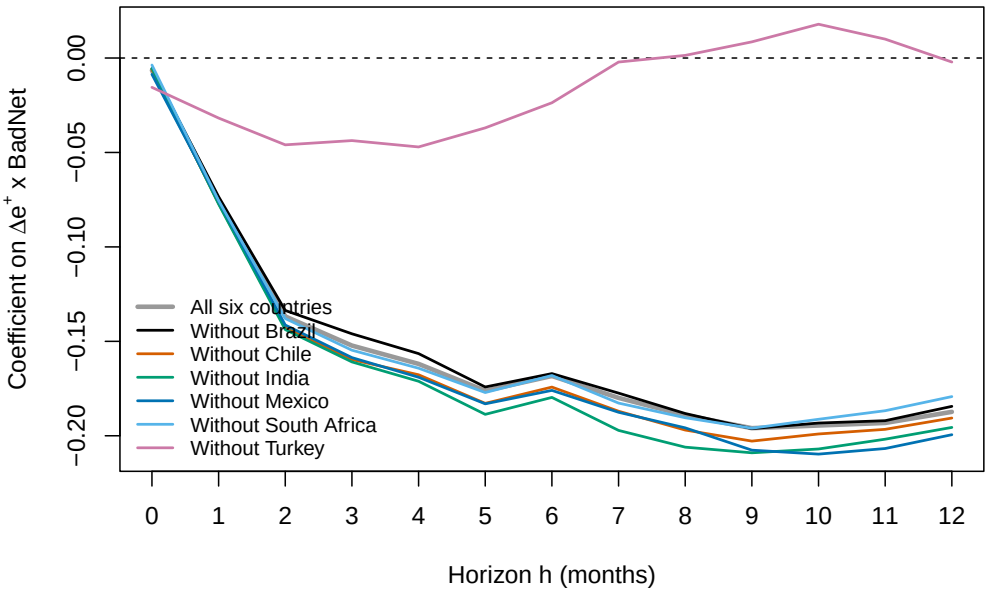


Figure A2: Leave-one-country-out estimates of the interaction

Note: Coefficient on $\Delta e^{+}_{i,t} \times BadNet_{i,t}$ in the baseline specification estimated without the country indicated, horizons 0 to 12. EM6 panel.

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